

Master's Thesis

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Markov Switching GARCH Models with Time-Varying Transition Distributions applied to High-Frequency Equity Returns

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Abstract

Markov Switching GARCH models, as specified by [Haas et al. \(2004\)](#), assume constant transition distributions. In this master's thesis, an implementation of Markov switching GARCH models with time-varying transition distributions is provided and implemented as an extension to the R package MSGARCH by [Ardia et al. \(2019\)](#). Subsequently, the model is applied to high-frequency (intraday) returns. Transition probabilities are related to past returns. This provides another way of observing the leverage effect in intraday financial return data. Additionally, a case study is conducted to compare the performance of multiple GARCH model specifications to assess the usefulness of time-varying transition probabilities and the applicability of this model to risk measurement and volatility modelling in financial institutions.

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1 Introduction

1.1 Literature Review

Markov switching models are non-linear time series models in which the progression of the time series is governed by states (or regimes) of the world. The realisations of these states are controlled by a hidden Markov chain with a finite state space. One of the earliest contributions was provided by [J. D. Hamilton \(1989\)](#); [J. D. Hamilton \(1988\)](#), who suggested to model non-stationary time series through a regime switching Markov model. Since the late 1980s, research in Markov switching models grew rapidly ([Guidolin, 2011](#)). The econometric applications of Markov switching models have become one of the main methods in the field of finance to facilitate the recovery of important (hidden) information about the state of the world through observed financial data. Moreover, Markov switching models are a popular tool in modelling various financial time series and are frequently employed by financial institutions ([Duprey et al., 2017](#)). For example, Markov switching models are used to model foreign exchange rates ([C. Engle et al., 1990](#); [Engel, 1994](#); [Dewachter, 2001](#); [Klaassen, 2002](#)), stock returns ([Schaller et al., 1997](#); [Rydén et al., 1998](#); [Maheu et al., 2000](#)) and return volatility ([Pagan et al., 1990](#); [J. D. Hamilton et al., 1994](#); [Gray, 1996](#); [Klaassen, 2002](#)).

This thesis focuses on the application of Markov switching models in the realm of volatility modelling. A pivotal approach to volatility models in financial markets was put forward by [R. F. Engle \(1982\)](#) and [Bollerslev \(1986\)](#), who introduced the (generalised) autoregressive conditional heteroskedasticity (G)ARCH model. Subsequently, researchers tried to introduce deterministic jumps into the volatility process. At first, [Cai \(1994\)](#) and [R. F. Engle \(1982\)](#) augmented the ARCH model with the Markov switching model, in which the variance process was governed by different volatility regimes. Researchers at the time were constrained to the ARCH specification of the more general GARCH model, as the *path dependence* problem made the model intractable (for details, we refer to [section 3](#)). Introducing multiple states into the GARCH model would demand knowledge of all possible trajectories across all possible states of the world, which made it difficult to estimate the model as the number of data points increased. However, [Gray \(1996\)](#) was able to construct a modification of the GARCH model to solve the dimensionality problem, which was subsequently adapted by [Klaassen \(2002\)](#). The most popular approach to overcome the path dependence problem is the parallel specification of the Markov switching GARCH (MSGARCH) model by [Haas et al. \(2004\)](#), which found its implementation in the prominent R package MSGARCH by [Ardia et al. \(2019\)](#).

In all preceding specifications and implementations of the MSGARCH model, the switching between states or regimes is governed by a Markov chain of first order. The probabilities that determine the transitions between states are commonly assumed to be constant. However,

Markov switching models with time-varying transition distributions provide a flexible framework for time series modelling, which are able to capture state dependent covariate effects with respect to state-transition probabilities. [Diebold et al. \(1994\)](#) were early proponents of Markov switching models with time-varying transition probabilities (TV-MSGARCH). Literature commonly suggests to model the transition probabilities by means of a multinomial logit or probit functional form. Through employing these functional forms, the transition probabilities can be parametrised through covariate vectors. These covariate vectors are allowed to change over time, which in turn renders the transition probabilities time-dependent. Even though there are some TV-MSGARCH specifications that follow the approach from [Gray \(1996\)](#) and [Klaassen \(2002\)](#) ([Brunetti et al., 2008](#); [Tan et al., 2021](#)) and there have been mixture models that incorporate time-varying mixing weights ([Haas et al., 2013](#)), it is apparent that the research with respect to MSGARCH models with time-varying transition distributions is rather scarce. Furthermore, at this point in time and to the best of our knowledge, there is no CRAN-package that provides an implementation of TV-MSGARCH models in R.

1.2 Research Question

It is of foremost interest to this master's thesis to model conditional volatility of equity returns by means of MSGARCH and to incorporate time-varying transition probabilities into the model. An implementation of TV-MSGARCH is provided as an extension to the R package MSGARCH. Subsequently, the model is applied to high-frequency intraday returns. As time-varying Markov switching is assumed to be dependent on covariates which are able to explain the transition distribution, a straightforward covariate choice presents itself in past returns. The asymmetric effect of returns onto volatility has been well documented, as reported by [R. F. Engle et al. \(2007\)](#), and has given sufficient reason in the past for it to be incorporated into volatility model specifications¹. Naturally, this leads to the question of whether similar effects can be observed through relating transition probabilities in the TV-MSGARCH model to past returns. Therefore, we pose the following research question: *"Do high-frequency (HF) returns have an effect on transition probabilities of volatility regimes and, if so, how do effects differ across the state space?"*. Ultimately, transition probabilities were related to past intraday returns. Our findings show that there is evidence that the leverage effect of volatility ([Black, 1976](#)) is present in intraday financial returns (observed thorough effects on transition probabilities). However, the evidence was not found to be strong.

¹For instance, the sign-switching GARCH model from [Fornari et al. \(1997\)](#) or the GJR-GARCH model provided by [Glosten et al. \(1993\)](#).

1.3 Outline

The thesis is structured in the following way. In [section 2](#), a basic introduction to Markov mixture modelling and Markov switching is provided. [Section 3](#) discusses Markov switching GARCH models, their estimation and implementation, with a special focus on the R package MSGARCH. [Section 4](#) provides an outline of how the aforementioned package was extended to incorporate time-varying transition probabilities and provides a walk-through of functionalities that are supported for TV-MSGARCH models. In [section 5](#), the extension of the package is used to apply the TV-MSGARCH model to high-frequency returns. In this chapter we try to answer the research question by observing estimated coefficients of past returns in the TV-MSGARCH model for multiple intraday return frequencies. Additionally, we test the out-of-sample performance of TV-MSGARCH on intraday and daily returns to assess whether this model provides added value in forecasting risk measures, such as Value-at-Risk, for financial institutions. [Section 6](#) provides a conclusive summary of our results.

2 Finite Markov Mixture Models for Time Series Modelling

In the interest of modeling time series by means of a Markov switching model (MSM), an understanding of basic properties of finite Markov mixture distributions has to be established first.

2.1 Preliminaries and Basic Definitions

The following chapter should provide a brief review on Markov chains and fundamental assumptions², which will let us advance to modelling time series through MSMs. For a detailed discussion, we refer the reader to [Frühwirth-Schnatter \(2006, Section 10, 11\)](#) and [J. Hamilton \(2020, Section 22\)](#).

Let $\{y_t\}_{t=1}^T$ denote a time series of T univariate observations, taking values in the continuous (or discrete) sampling space $\mathcal{Y}$. The observation y_t represents a realisation of the discrete-time stochastic process $\{Y_t\}_{t=1}^T$. In the theory of Markov mixture modelling it is assumed that the realisations of Y_t are governed by a hidden discrete-time stochastic process S_t . The process S_t is a discrete state indicator process that indicates the affiliation of y_t to a particular regime or state. The finite sample space of $\{S_t\}_{t=1}^T$ is an integer subset defined as $\mathcal{K} = \{1, \dots, K\}$. The realisations y_t are directly observable, however, the process $\{S_t\}_{t=1}^T$ is a latent random process and needs to be deduced by the effect it has on the distribution of Y_t . The process $\{S_t\}_{t=1}^T$ is assumed to be a Markov chain. The dynamics of the Markov chain are explained in the next section.

2.1.1 State Dynamics

MC1 MC2 Let S_t be a *homogeneous* Markov chain of *first order*. To elaborate, a Markov chain is called homogeneous if and only if the transition probabilities are independent of time – time-dependent probabilities will be discussed in later chapters (see [section 4](#)). A Markov chain of first-order is considered to be a chain for which the subsequent state is solely dependent on the preceding state. The state dynamics of S_t are described by the $K \times K$ transition probability matrix $\mathbf{P}$, which can be written as

$$\mathbf{P} = \begin{bmatrix} p_{1,1} & \dots & p_{1,K} \\ \vdots & \ddots & \vdots \\ p_{K,1} & \dots & p_{K,K} \end{bmatrix}.$$

Each element $P_{i,j}$ denotes the probability of transitioning from state i into state j for all $t = 1, \dots, T$, i.e.

$$P_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \quad \text{for all } j, i \in \{1, \dots, K\}.$$

²Assumptions regarding the Markov chain S_t are abbreviated with MC <<number>>.

The transition probabilities are non-negative and the row-sums of $\mathbf{P}$ are equal to one, i.e.

$$P_{ij} \geq 0, \forall j, i \in \{1, \dots, K\} \quad \text{and} \quad \sum_{j=1}^K P_{ij} = 1, \forall i = 1, \dots, K.$$

The basic theory of Markov mixture modelling puts some constraints on the Markov chain S_t . It is assumed that the latent state indicator process $\{S_t\}_{t=1}^T$ is an *irreducible* and *aperiodic* Markov chain.

MC3 MC4

Formally, a chain is irreducible if:

$$\exists n \in \mathbb{N} \forall i, j \in \{1, \dots, K\} \text{ s.t. } P_{i,j}^n > 0.$$

Note that $P_{i,j}^n$ denotes an element of the matrix product $\mathbf{P}^n = \mathbf{P} \cdots \mathbf{P}$ and represents the predictive probability $\Pr(S_{t+n} = i \mid S_t = j, \mathbf{P})$. Informally, irreducibility means that there is a finite sequence of transitions from any state to any other with positive probability. All states in the chain communicate with each other and every state in the chain can be reached from anywhere else in the chain. There are no absorbing states from which the chain cannot exit into another state.

An aperiodic Markov chain is defined to be a chain in which all states exhibit a *period*, defined as

$$d(k) := \gcd\{n \geq 1 : P_{k,k}^n\},$$

of 1, where $\gcd(\cdot)$ denotes the greatest common divisor. Essentially, the aperiodicity ensures that there are no regular intervals in which a chain returns to a particular state.

If a Markov chain is both, irreducible and aperiodic, it is called an *ergodic* Markov chain. Ergodic chains have the advantageous property of eventually reaching a unique stationary distribution. In the long-run, as shown by [J. Hamilton \(2020, Section 22.2\)](#), the probability of S_{t+n} (for $n \rightarrow \infty$), attaining a value in the state space $\mathcal{K}$, is independent of the current state at time t . As time progresses, it is important to know how frequent a state is (re-)visited. The unconditional probability $\Pr(S_t = k \mid \mathbf{P}) = \pi_k$ measures the fraction of times a chain spends in state k for all $k \in \mathcal{K}$. The corresponding distribution $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ denotes the invariant (or stationary) distribution of S_t . For general Markov chains the distribution $\boldsymbol{\pi}$ has to satisfy the *invariance property*

$$\mathbf{P}'\boldsymbol{\pi} = \boldsymbol{\pi} \tag{1}$$

in order to be called invariant distribution. If $\boldsymbol{\pi}$ satisfies [Equation 1](#), then $\Pr(S_t = k \mid \mathbf{P}) = \pi_k$

for all $k \in \mathcal{K}$ and $t = 0, 1, \dots, T$ (see Appendix A.1 for details).

Note that the invariant distribution that satisfies Equation 1 must not be unique wrt. the choice of $\mathbf{P}$. It is practical to work with ergodic Markov chains, since ergodic chains are ensured to have a unique invariant distribution (Frühwirth-Schnatter, 2006). Thus, for any irreducible and aperiodic Markov chain there exists a unique invariant distribution π , in other words, a solution to Equation 1. Due to the existence of a unique solution to Equation 1 under ergodicity, π_k are also referred to as *ergodic probabilities* (J. Hamilton, 2020; Frühwirth-Schnatter, 2006).

Ergodic probabilities provide information about the limiting distribution of S_t as it can be shown that $\lim_{n \rightarrow \infty} \mathbf{P}^n = \pi \mathbf{1}'$ for an ergodic Markov chain (J. Hamilton, 2020). To compute the ergodic distribution π , the system of linear equations, namely $\mathbf{P}'\pi = \pi$ and $\mathbf{1}'\pi = 1$, must be solved for π . The latter equation must hold as the probabilities of π must add up to one. As shown by J. Hamilton (2020), the problem can be reformulated as

$$\mathbf{A}\pi = \mathbf{e}_{K+1}, \quad \text{where} \quad \underset{(K+1) \times K}{\mathbf{A}} := \begin{bmatrix} \mathbf{I}_K - \mathbf{P}' \\ \mathbf{1}' \end{bmatrix}. \quad (2)$$

$\mathbf{I}_K$ is a $K \times K$ identity matrix and $\mathbf{e}_{K+1}$ is a $(K + 1)$ th standard basis vector. Multiplying the right hand side of the equation Equation 2 by $(\mathbf{A}\mathbf{A}')^{-1}\mathbf{A}'$ gives the solution to the linear system.

MC5

Ultimately, to ensure that a given Markov chain attains its ergodic distribution in the long-run, the initial probabilities of the chain need to be set according to the ergodic distribution, i.e. $\Pr(S_0 = k \mid \mathbf{P}) = \pi_k$ for all $k \in \mathcal{K}$. Hence, the *initial probabilities are assumed to follow the ergodic distribution*. In practice, for instance in the R package MSGARCH, the transition matrix $\mathbf{P}$ is estimated first and through Equation 2 the initial probabilities are derived to ensure that the Markov chain follows a stationary distribution. However, a different choice of starting distribution of S_t does not affect the identifiability of the Markov mixture model.

2.1.2 Dependence Structure

After discussing basic properties of homogeneous irreducible aperiodic Markov chains, we can advance to understand the dependence structure between Y_t and S_t in Markov mixture distributions. Subsequently, it will be possible to construct the basic Markov switching model.

The dependency of Y_t on S_t is given by the distributional assumption

$$Y_t \mid S_t = k \sim \mathcal{D}(\boldsymbol{\theta}_k), \quad (3)$$

where $\mathcal{D}$ is a parametric family of distributions over the same sampling space $\mathcal{Y}$ as mentioned in [subsection 2.1](#). The parameter $\boldsymbol{\theta}_k$ is dependent on the state k , which is indicated by its subscript. The random variable Y_t for $t \geq 0$ follows one particular distribution out of the family $\mathcal{D}(\boldsymbol{\theta}_1), \dots, \mathcal{D}(\boldsymbol{\theta}_K)$, dependent on the state indicator S_t . The random variables $Y_1, \dots, Y_T$ are independent conditional on knowing the entire trajectory of $\{S_t\}_{t=0}^T$. Given the aforementioned assumptions about the stochastic processes $\{Y_t\}_{t=1}^T$ and $\{S_t\}_{t=1}^T$, by the law of total probability, the marginal density of Y_t can be written as

$$p(y_t \mid \boldsymbol{\vartheta}) = \sum_{k=1}^K p(y_t \mid S_t = k, \boldsymbol{\vartheta}) \Pr(S_t = k \mid \boldsymbol{\vartheta}).$$

The distribution of Y_t is dependent on the realisation of S_t , i.e. given $S_t = k$, Y_t has density $p(y_t \mid \boldsymbol{\theta}_k)$. Consequently, as stated in [Frühwirth-Schnatter \(2006\)](#), referencing [Baum et al. \(1970\)](#), the unconditional distribution of Y_t can be expressed as a mixture distribution

$$p(y_t \mid \boldsymbol{\vartheta}) = \sum_{k=1}^K p(y_t \mid \boldsymbol{\theta}_k) \pi_k,$$

where the ergodic distribution $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ serves as a vector of mixing weights. Hence, the process $\{Y_t\}_{t=1}^T$ follows a so-called *finite Markov mixture* of the distributional family $\mathcal{D}(\boldsymbol{\theta})$.

2.2 Basic Markov Switching Model

The basic Markov switching model takes on a special form in line with [Equation 3](#). We assume a stochastic process $\{Y_t\}_{t=1}^T$, with realisations y_t , which are generated by a finite Markov mixture from the distributional family

$$Y_t \mid S_t \sim \mathcal{D}(\boldsymbol{\theta}_{S_t}), \quad (4)$$

where S_t is a hidden state homogeneous ergodic Markov chain that takes on values in the state space $\mathcal{K}$. The joint distribution of $\{Y_t\}_{t=1}^T$ and $\{S_t\}_{t=0}^T$ is equal to:

$$p(\mathbf{S}^T, \mathbf{y}^T) = p(S_0 \mid \boldsymbol{\vartheta}) \prod_{t=1}^T p(y_t \mid y^{t-1}, \mathbf{S}^t, \boldsymbol{\vartheta}) p(S_t \mid \mathbf{S}^{t-1}, \mathbf{y}^{t-1}, \boldsymbol{\vartheta}).$$

The density $p(y_t \mid \mathbf{y}^{t-1}, \mathbf{S}^t, \boldsymbol{\vartheta})$ represents the one-step ahead predictive density of Y_t conditional on knowing all observations $\mathbf{y}^{t-1} = (y_1, \dots, y_{t-1})$ and conditional on knowing all states $\mathbf{S}^t = (S_1, \dots, S_t)$ up to t . The probability mass function $p(S_t \mid \mathbf{S}^{t-1}, \mathbf{y}^{t-1}, \boldsymbol{\vartheta})$ represents the distribution

MC6

of S_t conditional on knowing all past realisations and states up to t . The variable ϑ denotes a set of known model parameters. This includes known parameters like the transition matrix $\mathbf{P}$. The assumptions made for the basic Markov switching model (MC1-6) can be quite limiting. For instance, assumption MC6 implies that the predictive density of Y_t is not allowed to depend on past states or observations and due to MC2, S_t is only dependent on the last state S_{t-1} i.e. $p(S_t|\mathbf{S}^{t-1}, \mathbf{y}^{t-1}, \vartheta) = p(S_t|S_{t-1}, \vartheta)$, which could pose another limitation. The next section discusses commonly used relaxation of assumptions made in the previous chapters, which are employed by researchers in the realm of MSMs.

2.3 More General Markov Switching Model

The assumptions made about the processes $\{Y_t\}_{t=1}^T$ and $\{S_t\}_{t=0}^T$ in the previous sections are sometimes rather restrictive. In the literature of Markov switching models some assumptions need to be relaxed, depending on the application ([Frühwirth-Schnatter, 2006](#), Section 10.3.4). For the purpose of building up to Markov switching GARCH models in subsequent sections, we state the relaxation needed to be able to work with this special class of MSMs. We assume for Y_t :

- The conditional distribution of Y_t given S_t can depend on past random variables $Y_1, \dots, Y_{t-1}$.
- The conditional density $p(y_t|\mathbf{y}^{t-1}, \mathbf{S}^t, \vartheta)$ is allowed to be dependent on exogenous covariates x_t and on past realisations $\mathbf{y}^{t-1}$.

In addition, the assumptions about the hidden Markov chain S_t can be relaxed. We assume:

- S_t can be an inhomogeneous first-order Markov chain³.
- The transition matrix does not have to be aperiodic or irreducible.
- The initial distribution $p(S_0|\vartheta)$ can be arbitrary. It does not have to be the ergodic distribution.
- The conditional distribution $p(S_t|\mathbf{S}^{t-1}, \mathbf{y}^{t-1})$ can be dependent on exogenous variables and lagged realisations $\mathbf{y}^{t-1}$.

2.4 Time-Varying Switching

With the relaxed assumptions made in the previous section, we are able to abandon assumption MC1 (if needed). Often homogeneous Markov chains are argued to be too restrictive as the switching probabilities cannot depend on underlying economical fundamentals as suggested by [Diebold et al. \(1994\)](#). Under MC1, transition probabilities cannot vary over the duration of the

³This assumption needs to be made in order to incorporate time-varying transition distributions later on.

time series and are forced to stay constant, even if the environment would suggest differently. [Filardo \(1994\)](#) was an early advocate of time-varying transition probabilities and applied them to analyse business cycles. The relaxation of homogeneity renders the construction of time-varying transition distributions possible. As a consequence, the transition matrix P_t is able to vary over time. The assumption that S_t is dependent on some exogenous random vector Z_t with observation vector z_t , dependent on time t , facilitates time-varying switching. In order to extract information about the state transition from an exogenous variable, one needs to define a model through which the transition probabilities P_t are modelled. One approach is to use the functional forms of *multinomial logistic regression* or *multinomial probit* models ([Frühwirth-Schnatter et al., 2019](#), Section 13) and parameterise the transition probabilities $P_{t,lk} = P(z_t, \gamma)$ through a covariate vector z_t and a vector of regression coefficients γ . Both the multinomial probit and logit models demand the specification of a reference state k_0 in order to be identifiable. For the logit version, the parametrised transition distribution is given by

$$P_{t,lk} = \frac{\exp(z_t' \gamma_{lk}^z + \gamma_{lk})}{\sum_{j=1}^K \exp(z_t' \gamma_{lj}^z + \gamma_{lj})} \quad \text{for all } l \in \mathcal{K}, \text{ and } k \in \mathcal{K} \setminus \{k_0\}. \quad (5)$$

The vector γ_{lk}^z in [Equation 5](#) corresponds to the *covariate effect* of z_t , whereas γ_{lk} represents the *independent effect* on the transition probability. For identification purposes γ_{lk_0} and $\gamma_{lk_0}^z$ are set to zero, which gives

$$P_{t,lk_0} = \frac{1}{1 + \sum_{j=1}^{K-1} \exp(z_t' \gamma_{lj}^z + \gamma_{lj})} \quad \text{for all } l \in \mathcal{K}.$$

Note that the transition distribution can also be modelled in a time-invariant way by means of the logit functional form by imposing the restriction $\gamma_{lk}^z = 0$ for all $l \in \mathcal{K}$ and $k \in \mathcal{K} \setminus \{k_0\}$. To make the coefficients γ^z comparable, it is assumed that the covariate vector z_t is normalised beforehand.

2.5 Estimation of Markov Switching Models

Given the assumptions made for general Markov switching models in [subsection 2.3](#), the observational density is defined as

$$p(y_t | \mathbf{y}^{t-1}, S_t, \Theta),$$

where Θ is the set of all unknown parameters. This includes all distribution, location and shape parameters, as well as unknown parameters that parameterise the transition matrix⁴.

⁴This includes γ for time-varying transition distributions.

The full likelihood for the observations $\mathbf{y} = \mathbf{y}^T$ reads:

$$\begin{aligned}
 p(\mathbf{y}|\Theta) &= \prod_{t=1}^T p(y_t|\mathbf{y}^{t-1}, \Theta) \\
 &= \prod_{t=1}^T \left(\sum_{k=1}^K p(y_t|\mathbf{y}^{t-1}, \Theta, S_t = k) \Pr(S_t = k|\mathbf{y}^{t-1}, \Theta) \right) \\
 &= \prod_{t=1}^T \left(\sum_{k=1}^K \sum_{l=1}^K p(y_t|\mathbf{y}^{t-1}, \Theta, S_t = k) \mathbf{P}_{t-1,lk} \Pr(S_{t-1} = l|\mathbf{y}^{t-1}, \Theta) \right), \quad (6)
 \end{aligned}$$

where $\Pr(S_{t-1} = l|\mathbf{y}^{t-1}, \Theta)$ and $\Pr(S_t = k|\mathbf{y}^{t-1}, \Theta)$ denote the *filtered probabilities* and the *one-step ahead predictions*, respectively. Both measures can be obtained through *Hamilton's Filter* (J. D. Hamilton, 1989), as described in J. Hamilton (2020, Section 22). The maximum likelihood estimation can be conducted numerically since the entire likelihood function can be expressed through the filter. Alternatively, the model can be estimated by means of the *Baum-Welch Algorithm* (Baum et al., 1970), which constitutes a special case of the EM algorithm (Dempster et al., 1977). Sometimes a combination of the two approaches is used. For instance, in the MSGARCH package the initial parameters are estimated through the Baum-Welch algorithm first and subsequently taken to initialise the numerical maximisation of the likelihood function. For the Bayesian estimation of the Markov switching model, the specification of a prior $p(\vartheta)$ is needed such that the posterior distribution can be written as:

$$p(\vartheta|\mathbf{y}) \propto p(\mathbf{y}|\vartheta) p(\vartheta).$$

The Bayesian approach utilises Markov chain Monte Carlo (MCMC) methods. The general approach, which is suggested by the literature (Frühwirth-Schnatter, 2006), employs techniques like data augmentation through treating the latent process S_t as missing data. In this setting it is possible to sample from the posterior distribution using Gibbs sampling.

3 Markov Switching GARCH Model

The literature on GARCH models suggests that the volatility process of stock returns is governed by volatility shocks and persistence in the volatility process. This led to the construction of the generalized autoregressive conditional heteroskedasticity model (GARCH) by [Bollerslev \(1986\)](#). To capture both, volatility- shocks and persistence, the basic GARCH model takes on the form

$$h_t = \alpha_0 + \alpha_1 \eta_{t-1}^2 \beta_1 h_{t-1} \quad \text{and} \quad \eta_t | \mathcal{F}_{t-1} \sim N(0, h_t), \quad (7)$$

where $\mathcal{F}_t$ represents the information up to time t . A measure of total persistence is the sum $\alpha_1 + \beta_1$. This measure, when estimated from daily stock market returns, is often close to one. In turn, this means that volatility shocks do not die out over time. It is often argued that the high persistence is due to shifts in unconditional variance of h_t , which can be observed in the estimated parameter α_0 . In a single-regime GARCH model this parameter is not able to shift over time. If one allows for shifts in the parameter α_0 , it can be shown that the persistence is reduced substantially ([Lamoureux et al., 1990](#)). To capture deterministic shifts in the variance process, [Lamoureux et al. \(1990\)](#) suggested to use Markov switching models.

Before the development of MSGARCH models, [Cai \(1994\)](#) and [J. D. Hamilton et al. \(1994\)](#) proposed a Markov switching ARCH model. The constraint to an ARCH volatility process had to be made as the direct translation of the GARCH model by [Bollerslev \(1986\)](#) to a Markov switching model would entail that the GARCH part of [Equation 7](#) would be dependent on the entire trajectory of states $\mathbf{S}^t$. This problem is known as the *path-dependence problem*. The problem becomes clearer if one expands the expression for h_t in [Equation 7](#) to

$$h_t = \sum_{i=0}^{t-1} \left[(\alpha_0(S_{t-i}) + \alpha_1(S_{t-i}) \eta_{t-1-i}^2) \prod_{j=0}^{i-1} \beta_1(S_{t-j}) \right] + h_0 \prod_{i=1}^t \beta_1(S_i),$$

which shows that h_t depends on the entire trajectory or path of $\{S_t\}$ up to time t . As the number of time points increases, the number of possible states K^T grows exponentially with the number of time points T . As a result, the estimation of the direct MSGARCH model becomes infeasible. [Gray \(1996\)](#) and [Klaassen \(2002\)](#) solved the path-dependence problem by altering the variance specification in the GARCH component of h_t to no longer depend on the full path. However, contrary to the approach proposed by the mentioned authors, [Haas et al. \(2004\)](#) introduce a variation of the variance specification in which a strict separation between high and low volatility regimes is incorporated. The regime variances are assumed to progress in parallel. It is argued that the strict separation between volatility regimes is a more intuitive approach than the one used by [Gray \(1996\)](#), which computes h_{t-1} as a weighted average of

regime variance at time $t - 1$, i.e. $h_{t-1} = \sum_{i=1}^K \Pr(S_{t-2} = i)h_{t-1,i}$. [Klaassen \(2002\)](#) follows a similar approach, but uses $\Pr(S_{t-1} = i)$, i.e. conditions on the information up to time $t - 1$ instead of $t - 2$. [Haas et al. \(2004\)](#) argue in favour of disaggregating the variance process into K parallel regimes due to the simplicity and analytical tractability of the approach. Perhaps, this is the reason why this approach was implemented by [Ardia et al. \(2019\)](#) in their implementation of the MSGARCH package.

The following sections discuss the MSGARCH model by [Haas et al. \(2004\)](#) and outline how [Ardia et al. \(2019\)](#) were able to generalise this approach to various different GARCH-type models.

3.1 Model Definition

First, we discuss the definition of [Haas et al., 2004](#) for the Markov switching GARCH model before we proceed with the more general model specification provided by [Ardia et al., 2019](#), which is able to incorporate a broader class of GARCH process specifications and distributional assumptions about the time series of interest.

As in previous sections, we let $\{Y_t\}_{t=1}^T$ be a univariate discrete-time stochastic process taking on values in the continuous sampling space $\mathcal{Y}$. The realisations of the process, $\{y_t\}$, satisfy

$$y_t = \eta_t h_{S_t,t}^{1/2}, \quad \text{where } \eta_t \stackrel{\text{iid}}{\sim} N(0, 1) \quad (8)$$

and $\{S_t\}_{t=0}^T$ is a homogeneous first-order irreducible Markov chain with its realisations drawn from the sample space $\mathcal{K} = \{1, \dots, K\}$. As per definition of [Haas et al., 2004](#), y_t is assumed to have zero mean and no serial correlation. Note that the process [8](#) can not have a state-specific mean μ_{S_t} , as it would cause the time series y_t to be auto-correlated. This would violate the GARCH assumption, which states that time series of innovations $y_t = r_t - \mu_t$, where $r_t - \mu_t$ is the de-meaned time series of returns, can not be auto-correlated and must follow a stationary white noise process.

The GARCH specification of [Haas et al. \(2004\)](#) solves the path dependence problem of Markov switching GARCH models through the assumption that the variance process progresses in K parallel regime processes. The innovations y_t follow the GARCH(1,1) model. The regime-dependent variance process $h_{S_t,t}$ reads

$$h_{S_t,t} = \alpha_{0,S_t} + \alpha_{1,S_t} y_{t-1}^2 + \beta_{1,S_t} h_{S_t,t-1}. \quad (9)$$

Under this specification the conditional variance h_t is a function of past data and the current

state S_t . Thus, there is no path dependence with respect to the latent state process and the path dependence problem can be overcome. As the variance process consists of K parallel regime processes, Equation 9 can be written in vector notation:

$$\begin{bmatrix} h_{1,t} \\ \vdots \\ h_{K,t} \end{bmatrix} = \begin{bmatrix} \alpha_1^0 \\ \vdots \\ \alpha_K^0 \end{bmatrix} + \begin{bmatrix} \alpha_1^1 \\ \vdots \\ \alpha_K^1 \end{bmatrix} y_{t-1}^2 + \begin{bmatrix} \beta_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \beta_K \end{bmatrix} \begin{bmatrix} h_{1,t-1} \\ \vdots \\ h_{K,t-1} \end{bmatrix}.$$

Given the invertibility condition $|\beta_k| < 1$ for all $k = 1, \dots, K$, Equation 9 can be expressed as an ARCH(∞) process (see Appendix A.2 for a derivation). The process reads:

$$h_t = [I - \text{diag}(\beta_1, \dots, \beta_K)]^{-1} \alpha_0 + \sum_{i=1}^{\infty} \text{diag}(\beta_1, \dots, \beta_K)^{i-1} \alpha_1 y_{t-i}^2.$$

From the components of h_t ,

$$h_{k,t} = \alpha_0(1 - \beta_k)^{-1} + \sum_{i=1}^{\infty} \beta_k^{i-1} \alpha_1 y_{t-i}^2,$$

it becomes apparent that the regime specific process is entirely dependent on the "within-regime" GARCH(1,1) parameters.

We now consider a separation between a low volatility regime and a high volatility regime. The specification chosen by Haas et al. (2004) has the advantage of posing a clear distinction between the two regime processes. In a scenario where a shift from a low volatility regime to a high volatility regime happens between $t - 1$ and t , the driving forces of the low volatility process, i.e. h_{t-1} and y_{t-1}^2 , will not be able to influence the variance of the high volatility regime h_t . The regime variance h_t is controlled right away by the high volatility regime. The proposition of this modelling approach is justified by the irregularities in return volatility, which are observed through very rapid and occasional shifts. It is unrealistic to expect that low (or high) levels of volatility will result in low (or high) volatility in the next period across both regimes. The persistence in the variance process shall be regime-specific, which is modelled through letting α_0 , α_1 and β depend on the concurrent state S_t , hence, the parallel specification in Equation 9.

3.2 More General Model Specifications

In Ardia et al. (2019)'s publication and in the corresponding R package, MSGARCH, the conditional variance process is defined as a broader class of GARCH(1,1) models. The general

specification for the time series reads

$$y_t = \eta_t h_{S_t,t}^{1/2} \sim \mathcal{D}(0, h_{S_t,t}, \Xi_{S_t}), \quad (10)$$

where $\mathcal{D}(0, h_{S_t,t}, \Xi_{S_t})$ is a continuous distribution with zero mean, variance $h_{S_t,t}$ and a vector of shape parameters Ξ_{S_t} . According to [Ardia et al. \(2019\)](#), referencing [McNeil et al. \(2015\)](#), the (conditional) mean of y_t is often assumed to be zero for high-frequency returns and is therefore deemed appropriate. The zero mean assumption, which is also made by [Haas et al. \(2004\)](#), does not pose any problem for a time series with non-zero mean as the time series can be corrected by de-meaning with a suitable conditional mean process.⁵ The Markov chain $\{S_t\}$ is assumed to have the same properties as in [subsection 3.1](#).

3.2.1 Variance Specifications

The approach, which is followed by [Haas et al. \(2004\)](#), defines the variance process as a regime-switching process comprised of K regimes, which advance in parallel through time. The parallel definition of the variance process provides a framework to extend the MSGARCH model of [Haas et al. \(2004\)](#) to other GARCH-type specifications to model the regime-specific variance. It is even possible to define multiple specifications for different regimes belonging to the same variance process. [Ardia et al. \(2019\)](#) define the variance process as a selection of the following GARCH-type models⁶:

The **ARCH** model in [Ardia et al. \(2019\)](#), introduced by [R. F. Engle \(1982\)](#), defines the regime-specific variance process as

$$h_{k,t} = \alpha_{0,k} + \alpha_{1,k} y_{t-1}^2,$$

which is a special form of the GARCH specification of [Bollerslev \(1986\)](#) used in [Haas et al., 2004](#). The parameter vector is given by $\theta_k = (\alpha_{0,k}, \alpha_{1,k})^T$, where $\alpha_{0,k} > 0$ and $\alpha_{1,k} \geq 0$ are needed for the process to be positive. To ensure stationary regime processes, the condition $\alpha_{k,1} < 1$ must be satisfied⁷.

The **GARCH** model, introduced by [Bollerslev \(1986\)](#), follows the known model by [Haas et al. \(2004\)](#) (see [subsection 3.1](#)). The parameter vector is given by $\theta_k = (\alpha_{0,k}, \alpha_{1,k}, \beta_{1,k})^T$. Positivity of the process demands $\alpha_{0,k} > 0$, $\alpha_{1,k} > 0$ and $\beta_{1,k} \geq 0$. Stationarity of the variance process requires $\alpha_{1,k} + \beta_{1,k} < 1$.

⁵That being said, it is not possible to jointly estimate mean and volatility in the MSGARCH package.

⁶The GARCH specifications are implemented in the R package MSGARCH.

⁷Note that all positivity and stationarity conditions must hold for all regimes k with the corresponding GARCH specification. This applies to all other GARCH-type specifications as well.

The exponential GARCH or **E-GARCH** model in [Ardia et al. \(2019\)](#), introduced by [Nelson \(1991\)](#), defines the regime-specific variance process as

$$\log(h_{k,t}) = \alpha_{0,k} + \alpha_{1,k}(|\eta_{k,t-1}| - \mathbb{E}_k(\eta_{k,t-1})) + \alpha_{2,k}\eta_{k,t-1} + \beta_{1,k}\log(h_{k,t-1}),$$

where $\mathbb{E}_k$ indicates that the expectation is taken with respect to the regime-specific distribution of $\eta_{k,t-1}$. The parameter vector is given by $\theta_k = (\alpha_{0,k}, \alpha_{1,k}, \alpha_{2,k}, \beta_{1,k})^T$. Positivity of $h_{k,t}$ is ensured per definition. Stationarity of the regime-specific variance process requires $\beta_{1,k} < 1$. Note, the model specification is an asymmetric version of GARCH-type processes as negative shocks have a higher influence on the regime-variance than positive shocks. This can be observed as the parameters $\alpha_{2,k}$ and $\alpha_{1,k}$ are most often estimated to be negative and positive, respectively. The asymmetric specification is supposed to capture the *leverage effect* of volatility, which is a stylized fact about return volatility, under which negative past returns are ought to predict higher volatility in subsequent periods ([Black, 1976](#); [R. F. Engle et al., 2007](#)).

The **GJR-GARCH** model, introduced by [Glosten et al. \(1993\)](#), is another asymmetric variation of the GARCH model by [Bollerslev \(1986\)](#). The regime-specific volatility process in [Ardia et al. \(2019\)](#) is defined as

$$h_{k,t} = \alpha_{0,k} + (\alpha_{1,k} + \alpha_{2,k}\mathbb{I}_{y_{t-1} < 0})y_{t-1}^2 + \beta_{1,k}h_{k,t-1},$$

where $\mathbb{I}_{\{\cdot\}}$ is the indicator function, which takes on the value one if the return at time $t - 1$ is negative and, else, is zero. The model is a linear piecewise function of past returns. Through the separation of negative and positive return effects, the leverage effect is embedded into the model. The parameter vector is given by $\theta_k = (\alpha_{0,k}, \alpha_{1,k}, \alpha_{2,k}, \beta_{1,k})^T$. Positivity of the process is ensured through $\alpha_{0,k} > 0, \alpha_{1,k} > 0, \alpha_{2,k} \geq 0$ and $\beta_{1,k} \geq 0$. Stationarity of the regime-specific variance process, according to [Ardia et al. \(2019\)](#), referencing [Zakoian \(1994\)](#), requires the condition $\alpha_{1,k} + \alpha_{2,k}\mathbb{E}(\eta_{k,t}^2\mathbb{I}_{\eta_{k,t} < 0}) + \beta_{1,k} < 1$.

The **T-GARCH** model, introduced by [Zakoian \(1994\)](#), models the square-root of the regime-specific variance process. It poses another asymmetric GARCH specification and reads

$$h_{k,t}^{1/2} = \alpha_{0,k} + (\alpha_{1,k}\mathbb{I}_{y_{t-1} \geq 0} - \alpha_{2,k} - \alpha_{2,k}\mathbb{I}_{y_{t-1} < 0})y_{t-1} + \beta_{1,k}h_{k,t-1}^{1/2}.$$

The parameter vector is given by $\theta_k = (\alpha_{0,k}, \alpha_{1,k}, \alpha_{2,k}, \beta_{1,k})^T$. Positivity is ensured through the conditions $\alpha_{0,k} > 0, \alpha_{1,k} > 0, \alpha_{2,k} > 0$ and $\beta_{1,k} \geq 0$. Stationarity of $h_{k,t}$ is ensured by $\alpha_{1,k}^2 + \beta_{1,k}^2 - 2\beta_{1,k}(\alpha_{1,k} + \alpha_{2,k})\mathbb{E}(\mathbb{I}_{\eta_{k,t} < 0}) - (\alpha_{1,k}^2 - \alpha_{2,k}^2)\mathbb{E}(\eta_{k,t}^2\mathbb{I}_{\eta_{k,t} < 0}) < 1$.

Stationarity and positivity conditions throughout all variance specifications are ensured through

parameter-mappings in the MSGARCH package. A common practice is to use the inverse generalised logit function to map parameters from the domain $(-\infty, +\infty)$ to $[\min, \max]$, where $\min$ and $\max$ denote model-specific bounds. The function reads

$$f(x) = \min + \frac{\max - \min}{1 + \exp(-x)} \in [\min, \max].$$

Through the inverse of $f()$, parameters can be unmapped again. This approach is used in the numerical optimisation step of the MSGARCH package, where numerically optimised parameters are ensured to stay within their respective bounds such that positivity and stationary conditions are fulfilled at all times.

3.2.2 Distributions of Innovations

The assumed distributions of innovations $\eta_{k,t}$ can influence the performance of the MSGARCH model. There are three main distributions which are suggested by [Ardia et al. \(2019\)](#) and provided in the MSGARCH package. The given distributions are popular distributions to model log-returns in financial markets. The distributions to choose from are the *standard Normal*, the *Student-t* and the *Standardised Generalised Error* (GED) distribution. The shape parameter ν_k of the Student-t and GED distribution add an additional parameter to the set of model parameters that need to be estimated. Through the *Fernandez-and-Steel transformation*, implemented by [Ardia et al. \(2019\)](#), introduced by [Fernández et al. \(1998\)](#), it is possible to construct skewed versions for the aforementioned distributions, which makes a total of six different distributions that are supported by the R package. The skewed versions of the three distributions add an additional "skewness" parameter $\xi_k \in (0, 1]$ to the set of model parameters. The GJR, EGARCH and TGARCH models depend on first and second moments of $\eta_{k,t}$. [Trottier et al. \(2016\)](#) derive such moments for standardised Fernandez-and-Steel transformed distributions to facilitate estimation. Ultimately, the parameters ν_k and ξ_k are both shape parameters and are summarised in $\Xi_k = (\nu_k, \xi_k)$ ⁸.

3.3 Estimation

The estimation of the MSGARCH model by [Ardia et al. \(2019\)](#) can be conducted through Maximum likelihood or Bayesian Markov Chain Monte Carlo (MCMC) estimation techniques. We are going to discuss the maximum likelihood estimation exclusively as it is relevant to time-varying transition distribution in the extension of the MSGARCH package, which is discussed in later chapters (see [section 4](#)). For an outline of the MCMC estimation technique we refer to [Ardia et al. \(2019\)](#). In any case, one needs to derive the likelihood function for the model,

⁸The notation is not entirely correct since it is possible that for any $k \in \mathcal{K}$ one or neither one of the parameters (ν_k, ξ_k) is needed. Thus, technically this is not a vector, however, the notation is kept for simplicity and readability.

and the corresponding data, to facilitate estimation. The likelihood function is dependent on a set of previously mentioned parameter vectors. We summarise variance model-specific parameters $(\theta_1, \dots, \theta_K)$, shape parameters $(\Xi_1, \dots, \Xi_K)$ and the transition probability matrix P in the parameter set $\Theta = (\theta_1, \Xi_1, \dots, \theta_K, \Xi_K, P)$.

The likelihood function for the Markov switching model of interest is:

$$p(\mathbf{y}|\Theta) = \prod_{t=1}^T f_{\mathcal{D}}(y_t|\Theta, \mathcal{F}_t),$$

which, by the general result for MSMs in Equation 6, equates to

$$p(\mathbf{y}|\Theta) = \prod_{t=1}^T \left(\sum_{k=1}^K \sum_{l=1}^K f_{\mathcal{D}}(y_t|\mathcal{F}_{t-1}, \Theta, S_t = k) P_{t-1,lk} \Pr(S_{t-1} = l|\mathcal{F}_{t-1}, \Theta) \right),$$

where $P_{t-1,lk} = \Pr(S_{t-1} = k|S_{t-2} = l, \mathcal{F}_{t-2}, \Theta)$ and $\mathcal{F}_t$ represents the information with respect to the observations of time series $\{y_t\}$ and $\{z_t\}$ up to time t . Note that [Ardia et al. \(2019\)](#) do not include time-varying transition probabilities. Thus, the information set $\mathcal{F}_t$ does not include any time series observations $\{z_t\}$. The expression $f_{\mathcal{D}}(\cdot|\mathcal{F}_{t-1}, \Theta, S_t = k)$ denotes the probability density function of $\mathcal{D}$, conditional on the information $\mathcal{F}_{t-1}$, Θ and the Markov chain S_t being in state k at time t . The term $\Pr(S_{t-1} = l|\mathcal{F}_{t-1}, \Theta)$ denotes the filtered state probability.

3.3.1 Maximum Likelihood Estimation

The maximum likelihood (ML) estimation, as already established in [subsection 2.5](#), can be conducted through the Baum-Welch algorithm, numerically or through a combination of the two approaches. In the MSGARCH package, i.e. in the implementation of [Ardia et al. \(2019\)](#), the ML estimator of Θ is obtained through the following algorithm:

Initially the starting values for Θ are chosen. This procedure entails estimating the static version of the model, i.e. treating the observations of the variance process $h_{S_t,t} = \bar{h}_{S_t}$ as given and applying the *Baum-Welch algorithm* ([Baum et al., 1970](#)) to estimate starting values for the transition matrix P . At this point, it is assumed that the Markov chain S_t is homogeneous and that the transition matrix is time-independent.

Baum-Welch

The Baum-Welch algorithm, as a variation of the expectation maximisation (EM) algorithm, tries to iteratively find a maximum likelihood estimate in models containing hidden data (states/regimes). It utilises the *Forward-Backward* algorithm to estimate the quantities needed

for the expectation step. The parameters that are to be estimated are $\vartheta = (\mathbf{A}, \mathbf{B}, \boldsymbol{\pi})$, where

- the elements of $\mathbf{A} \in [0, 1]^{K \times K}$ are transition probabilities $a_{ij} = \Pr(S_t = j | S_{t-1} = i)$,
- the elements of $\boldsymbol{\pi} \in [0, 1]^K$ are ergodic probabilities $\pi_i = \Pr(S_1 = i)$ and
- the elements of $\mathbf{B} \in \mathbb{R}_+^{T \times K}$ are $b_j(y_t) = \Pr(Y_t = y_t | S_t = j) = f_N(y_t | S_t = j)$, i.e. conditional densities for the observation y_t on S_t being in state j under the assumption that Y_t follows a normal distribution.

The initial values, before setting of the EM algorithm, for the diagonal entries of $\mathbf{P}$ are set to 0.9 and the off-diagonal entries are set to be the same such that the row-sums are equal one. The initial values of $\boldsymbol{\pi}$ are set according to the invariance property as previously discussed in Equation 2. The initial values for the component-specific variance $\sigma_{S_t}^2$ and mean μ_{S_t} are estimated by scaling the sample variance of the observation vector $\mathbf{y}$ by an increasing factor between 0.4 and 1. Ultimately, the goal is to find

$$\vartheta_{MLE} = \arg \max_{\vartheta} p(\mathbf{y} | \vartheta).$$

In the Baum-Welch algorithm there are two key quantities which are of interest:

- $\alpha_i(t) = \Pr(Y_1 = y_1, \dots, Y_t = y_t, S_t = i | \vartheta)$,
- $\beta_i(t) = \Pr(T_{t+1} = y_{t+1}, \dots, Y_T = y_T | S_t = i, \vartheta)$.

Both measures are computed iteratively for $t = 1, \dots, T$ and $i \in \mathcal{K}$ through the *Forward Procedure* and *Backward Procedure*, respectively. The Forward-Backward algorithm follows the following procedures:

The Forward procedure is given by

$$\begin{aligned} 1) \quad & \alpha_i(1) = \pi_i b_i(y_1), \\ 2) \quad & \alpha_i(t+1) = b_i(y_{t+1}) \sum_{j=1}^K \alpha_j(t) a_{ji}, \end{aligned}$$

and the Backward procedure follows

$$\begin{aligned} 1) \quad & \beta_i(T) = 1, \\ 2) \quad & \sum_{j=1}^K \beta_j(t+1) a_{ij} b_j(y_{t+1}). \end{aligned}$$

Through the combination of both measures, $\alpha_i(t)$ and $\beta_i(t)$, we get

$$p(\mathbf{y}|\vartheta) = \sum_{i=1}^K \alpha_i(t) \beta_i(t) = \sum_{i=1}^K \Pr(\mathbf{y}, S_t = i|\vartheta).$$

Subsequently, we are able to compute the smoothed probability $\gamma_i(t)$ of being in state i at time t , given the entire trajectory $\mathbf{y}$ and some parameter set ϑ , by means of Bayes' Theorem:

$$\gamma_i(t) = \Pr(S_t = i|\mathbf{y}, \vartheta) = \frac{\Pr(S_t = i, \mathbf{y}|\vartheta)}{p(\mathbf{y}|\vartheta)} = \frac{\alpha_i(t) \beta_i(t)}{\sum_{j=1}^K \alpha_j(t) \beta_j(t)}.$$

In addition, the probability $\delta_{ij}(t)$ is derived as

$$\delta_{ij}(t) = \Pr(S_t = i, S_{t+1} = j|\mathbf{y}, \vartheta) = \frac{\Pr(S_t = i, S_{t+1} = j, \mathbf{y}|\vartheta)}{p(\mathbf{y}|\vartheta)} = \frac{\alpha_i(t) a_{ij} \beta_j(t+1) b_j(y_{t+1})}{\sum_{k=1}^K \sum_{w=1}^K \alpha_k(t) a_{kw} \beta_w(t+1) b_w(y_{t+1})},$$

which is the probability of the transition between S_t and S_{t+1} at time t , given the entire trajectory of $\mathbf{y}$ and some parameter set ϑ .

In the *expectation step* the expectations for the number of transitions are estimated. The expected number of transitions into the state i is estimated through $\sum_{t=1}^{T-1} \gamma_i(t)$. The expected number of transitions from state i to state j is estimated to be $\sum_{t=1}^{T-1} \delta_{ij}(t)$.

In the *maximisation step* the parameters in ϑ are re-evaluated as follows:

$$\pi_i^* = \gamma_i(1) \quad \text{and} \quad a_{ij}^* = \frac{\sum_{t=1}^{T-1} \delta_{ij}(t)}{\sum_{t=1}^{T-1} \gamma_i(t)}.$$

As the normal distribution is completely defined through mean μ_{S_t} and variance $\sigma_{S_t}^2$, instead of updating values for the emission matrix $\mathbf{B}$, only the parameters μ_{S_t} and σ_{S_t} are updated in MSGARCH. Hence,

$$\mu_i^* = \frac{\sum_{t=1}^T \gamma_i(t) y_t}{\sum_{t=1}^T \gamma_i(t)} \quad \text{and} \quad \sigma_i^{2*} = \frac{\sum_{t=1}^T \gamma_i(t) (y_t - \mu_i^*)^2}{\sum_{t=1}^T \gamma_i(t)}.$$

The previous steps are repeated until a convergence criterion is met, which, in MSGARCH, is measured as a relative difference between log-likelihood values. At this point it is worth noting that [Ardia et al. \(2019\)](#) implemented the Baum-Welch algorithm using the log-likelihood of the data. In addition, the Forward-Backward algorithm is implemented with a scaling scheme, introduced by [Rabiner \(1989\)](#), to avoid underflow through multiplication of many probability

measures that are significantly smaller than one.

An advantage of the Baum-Welch algorithm is that at each step one is guaranteed $p(\mathbf{y}|\theta_{k+1}) \geq p(\mathbf{y}|\theta_k)$. However, the algorithm is not guaranteed to converge to a global optimum. Another drawback of the EM algorithm is that it can be slow to converge. This is presumably the reason why [Ardia et al. \(2019\)](#) use a second numerical optimisation step after solely estimating the starting parameters with the EM algorithm.

After estimating the initial matrix $\mathbf{P}$, [Ardia et al. \(2019\)](#) use the *Viterbi algorithm* ([Viterbi, 1967](#)) to assign each observation a regime. Subsequently, the regime-specific observations are collected in K different time series. For each time series a single-regime GARCH-type model is fitted through quasi-maximum likelihood estimation to estimate all remaining parameters of Θ . The quasi-maximum likelihood estimation is done through numerically maximising the log-likelihood function of the respective time series. The algorithm that is used by [Ardia et al. \(2019\)](#) for the maximisation⁹ is the BFGS algorithm ([Broyden, 1970](#)), which is a quasi-Newton procedure in which the Hessian of the log-likelihood function is approximated.

Ultimately, the initial parameters for Θ are provided to optimise the log-likelihood of the complete MSGARCH model. To obtain the log-likelihood is not trivial and straightforward optimisation is not feasible. The log-likelihood function $\log p(\mathbf{y}|\Theta)$, as established in [subsection 2.5](#), is obtained through Hamilton's filter and optimised through the BFGS algorithm. The resulting parameters represent the final output of the ML estimation procedure in MSGARCH.

One of the main contributions of this thesis is the extension of the MSGARCH package with time-varying transition distributions, i.e. to support the TV-MSGARCH model. The subsequent section explains how and which changes were made to the source code of the package. Additionally, an outline of new functionalities is provided by means of R code examples.

⁹In MSGARCH it is a minimisation of the negative log-likelihood.

4 Time-varying MSGARCH

Haas et al. (2004) argue against employing time-varying transition probabilities as discussed in subsection 2.4 since the regime variances, as a result, would not be constant over time. The authors state that the variance process in the GARCH setting is supposed to be governed by shocks and not by regime probabilities. Furthermore, it was stated that the economic significance of incorporating time-varying switching is not clear. However, since the publication in 2004 there has been some evidence that time-varying regime switching could provide better out-of-sample performance for certain applications, as shown by Wang et al. (2021) and Tan et al. (2021).

One of the main contributions of this thesis is the implementation of time-varying transition probabilities in the MSGARCH package. The package is arguably the most prominent R package to provide an implementation of Markov switching GARCH models. Thus, instead of providing a separate package, we settled on the fact that the direct extension of the existing package would be a more valuable contribution. Through this approach we are able to build on the already existing code base and end-users only have to consider one single implementation when estimating MSGARCH models in practice. The extension of MSGARCH focuses only the maximum likelihood estimation methods provided in the original package, as extending the Bayesian estimation procedure is not feasible due to the time-constraints associated with this thesis.

In the TV-MSGARCH model, transition probabilities are estimated through an appropriate functional form as outlined in subsection 2.4. We choose to model the transition probabilities through the multinomial logit functional form. For the ML estimation of the TV-MSGARCH model, exogenous covariate matrices must be provided to the MLE method. The estimation procedure follows the same approach as outlined in subsubsection 3.3.1. That entails estimating the starting parameters of the model under the assumption of a homogeneous Markov chain S_t . However, the numerical maximisation of the log-likelihood function, by means of Hamilton's filter and the BFGS algorithm, is conducted by incorporating time-varying transition distributions. The estimation is analogous to constant transition distributions with the technical difference that the BFGS algorithm no longer finds the optimal P , among other parameters in Θ , but the vector of coefficients γ_{lk}^z and γ_{lk} . Essentially, the time-varying parameterisation of P , as in Equation 5, would imply exchanging P in Θ for the set of coefficients

$$\Gamma := \{ (\gamma_{lk}, \gamma_{lk}^z) \mid \text{for all } l \in \mathcal{K} \text{ and } k \in \mathcal{K} \setminus \{k_0\} \}.$$

In the subsequent section the extension of the R package MSGARCH is discussed. It gives an outline of how the object-oriented programming approach, which is employed in the package, was utilised to extend the package.

4.1 Extending the MSGARCH Package

The R package MSGARCH is primarily built on the packages Rcpp (Eddelbuettel et al., 2011) and RcppArmadillo (Eddelbuettel et al., 2014) as these packages provide an interface to C/C++ code to R. Computational intensive tasks and algorithms are implemented through C/C++ code, which makes the estimation of computational intensive models like MSGARCH or TV-MSGARCH more efficient. The R user only gets to provide the right parameters to certain R functions and retrieves the final output. The user does not get to interact with the source code itself. Everything is encapsulated in the R environment.

Nonetheless, in order to extend the package for time-varying switching, one needs to adapt the source code itself. The source code of the package follows an object-oriented programming paradigm, i.e. classes with member- data and functions are created to represent objects like a single-regime model, a MSGARCH model, a distribution or a particular GARCH-specifications, etc.. Throughout the runtime of a function call, the states of these class-objects are changed to generate the final output. The concept of *composition* stands at the center of the source code as the main class, MSgarch, which is later on exposed to the R environment, is a composition of K SingleRegime objects. In this context "exposed" means that through Rcpp *modules* entire classes, written in C++, and their member data and member functions can be exported/defined in an R object (Eddelbuettel et al., 2022). Subsequently, it is possible to use all functionalities of the C++ class from within the R environment.

The class MSgarch contains a vector of memory pointers to SingleRegime objects, which is how the composition is achieved. The SingleRegime class itself inherits from the class Base. The Base class is a base *virtual* class. All its members are virtual, meaning that all members can be over-written by the child class SingleRegime and the child class itself is still able to be used with parent class memory pointers. This means that we can use parent class pointers and call methods of any child class without even knowing the kind of derived class object, which is useful since the class SingleRegimes can have various types as it is a *templated* class. SingleRegime being templated class means that the class behaviour changes depending on the class that is provided as Model, as shown in the code-chunk below.

```
template <typename Model>
class SingleRegime : public Base {
    Model spec;
```

```
...
}
```

The class of `Model` can be one of the following: `sARCH<distribution>`, `sGARCH<distribution>`, `tGARCH<distribution>`, `eGARCH<distribution>` and `gjrGARCH<distribution>`.

The `<distribution>` denotes a viable distribution class, which is also implemented as a skewed- or symmetric templated class corresponding to the choices available in [subsubsection 3.2.2](#). For example, this could be a GARCH model with normal innovations, i.e. a `sGARCH< Symmetric<Normal> >` object or `sGARCH< <Skewed<Normal> >`, depending on whether skewed distributions are to be assumed.

The class `MSgarch` has various member functions associated with itself, which help to estimate the model through, both, the MLE and the Bayesian MCMC approach. Thereby, it is assumed that the transition matrix stays constant. To incorporate time-varying switching probabilities, we introduce a new `TVMSgarch` class that inherits all the base behavior from the `MSgarch` class. Additionally, a member function to compute transition distributions, given a vector of covariates, is added to the new class. The member function, essentially, makes use of the multinomial logit function and computes all components of the transition matrix based of the observation vector. The member function which were inherited by `MSgarch` are re-written such that constant transition matrix objects are replace by the parameterised version. For instance, this included the `HamiltonFilter` function that is implemented with constant transition distributions in the parent class – it needed to be over-written in the child class `TVMSgarch` to render the ML estimation for time-varying switching possible. An outline of the class and its relation to other classes is provided in the diagram below.

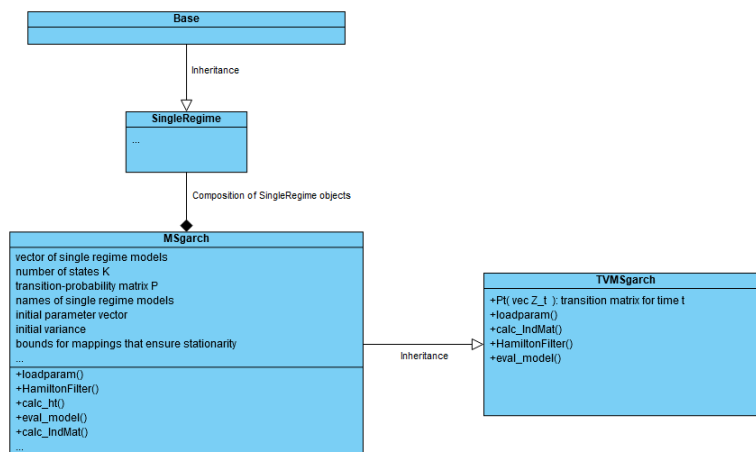


Figure 1: A preview on classes in the source code of the package MSGARCH

Changes are made to the parts of the code that originally initialised the `MSgarch` class. Depending on whether the user wants to construct a MSGARCH or TV-MSGARCH model (through the method `CreateSpec()`), the C++ class `MSgarch` or `TVMSgarch` are initialised in the background during runtime.

There are some changes made to the rest of the MSGARCH package, which mainly focus on connecting the newly written C++ class with rest of the code base, which is written in R. We are not going to discuss these changes further, as they pose no new feature to the package. These changes are merely necessary to make already present functionalities work for the TV-MSGARCH model and to integrate the new source code in the rest of the package in a fluent way. The entire code base for the extended package can be found online on <https://github.com/mkaywins/MSGARCH>. The following section outlines the functionalities which are supported for the TV-MSGARCH model.

4.2 Time-Varying Switching in the MSGARCH Package

This section provides a walk-through of functionalities, related to TV-MSGARCH models, which are supported in the extended MSGARCH package.

4.2.1 Installation

The extensions made to the original package are not available through the package on the official CRAN repository. In order to use the extension, one needs to download the forked¹⁰ package through the aforementioned GitHub page. This can be done with the help of the `devtools` package, which should be installed first.

```
# Download copy of the package from GitHub:
devtools::install_github("mkaywins/MSGARCH", subdir="Package")
```

After downloading, the package can be loaded into the R environment.

```
library(MSGARCH)
```

4.2.2 Maximum Likelihood Estimation

Before we fit a (TV-)MSGARCH model to data, we need to define the model specification through `CreateSpec`. This step is analogous to the basic version of the package with the key difference that in the `switch.spec` argument the variable `do.tvp` must be set to `TRUE`.

¹⁰A fork is a copy of a repository that belongs to a different account.

```
# To fit the model with time-varying probabilities (TVP), we must set
# do.tvp = TRUE:
spec = CreateSpec(switch.spec = list(do.tvp=TRUE),
                  variance.spec = list(model = c("sGARCH", "sGARCH")),
                  distribution.spec = list(distribution = c("norm", "norm")))
```

From the output of the object `spec` we can retrieve some basic information about the model specifications that we have chosen.

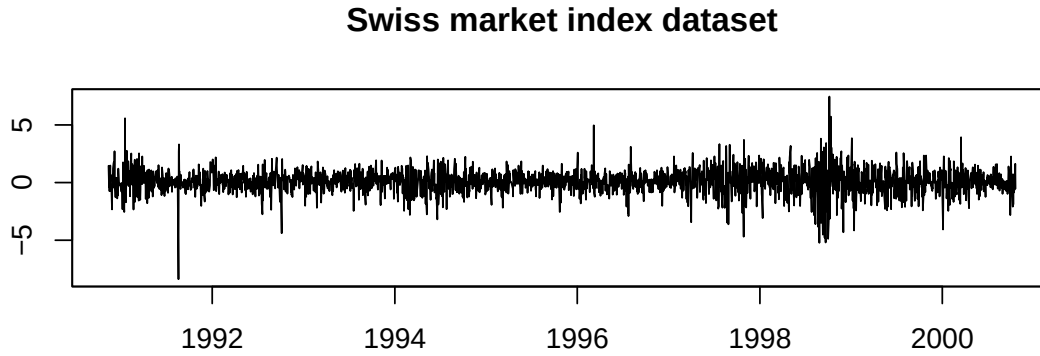
```
print(spec)

## Specification type: TV Markov-switching
## Specification name: sGARCH_norm sGARCH_norm
## Number of parameters in each variance model: 3 3
## Number of parameters in each distribution: 0 0
## -----
## Fixed parameters:
## None
## -----
## Across regime constrained parameters:
## None
## -----
```

Time-Varying switching is supported for all supported variance processes and distributions of innovations mentioned in [Ardia et al. \(2019\)](#). However, it is not possible at this point to use mixture-effects, which is a feature only supported for constant transition distributions. This means that the mixture analog to TV-MSGARCH ([Haas et al., 2013](#)) is not supported by the extended package at this point in time.

The base package comes with a test data set of log-returns of the Swiss stock market index from 1990 to 2000. We use this data set to fit a TV-MSGARCH model in the following.

```
# test data
data("SMI", package = "MSGARCH")
data = SMI
plot(data, type = "l", xlab="", ylab="", main = "Swiss market index dataset")
```



We model the time-varying transition probabilities by means of the multinomial logit functions as per [Equation 5](#). Similar to [Haas et al. \(2013\)](#), who constructed Gaussian mixture GARCH models by modelling time-varying mixing-weights through lagged returns, we choose the covariate matrix $Z \in \mathbb{R}^{N \times 2}$ to be

$$\begin{bmatrix} 1 & \text{lag}(y_1) \\ \vdots & \vdots \\ 1 & \text{lag}(y_T) \end{bmatrix}.$$

Through this parameterisation of the transition distribution, different probabilities can be estimated as the signs of lagged returns change. This could then be interpreted as a different representation of the *leverage effect* ([Black, 1976](#)) or Engle's news impact curve as suggested by [Haas et al. \(2013\)](#).

```
# Constructing the covariate matrix Z
Z = matrix(1, nrow = length(data)-1, ncol = 2)
Z[,2] = data[1:2499]
data = data[2:2500]
```

Subsequently, we fit the model through the MLE method `FitML`. Note that the covariate matrix Z only needs to be supplied for time-varying switching. Thus, it only works if the object `spec` includes the correct specifications (`do.tvp = TRUE` and `do.mix = FALSE`).

```
# We fit the model via the maximum likelihood method:
fit = FitML(spec = spec, data = data, Z = Z)
```

We can output a summary of the estimated model and statistics.

```
summary(fit)
```

The full output can be found in Appendix A.3. Appendix A.3 also shows that the TV-MSGARCH model, when compared with a MSGARCH model, can achieve a better in-sample fit. This is not a behavior that we choose to investigate further in this thesis, however, it shows that TV-MSGARCH can actually achieve a relatively good in-sample fit. The out-of-sample performance and how the model compares to other GARCH-type models is investigated in section 5.

4.2.3 State Information and Visualisations

By applying the function `State()` to the `fit` object from the previous computations, it is possible to retrieve information about filtered, smoothed and predictive probabilities, which are the probability measures that arise in the Forward-Backward algorithm. All probabilities are computed through time-varying switching probabilities. In addition, it is also possible to extract the estimated states that are estimated through the Viterbi algorithm. The object `state` in the code below can then be used with the generic print function and the additional argument `type.prob` to plot the time series of aforementioned probabilities and estimated states.

```
# Extract fitted states, filtered, smoothed, predictive probabilities
state = State(fit)

op = par(mfrow = c(3,3))
plot(state, type.prob = "smoothed") # smoothed probs
plot(state, type.prob = "predictive") # predictive probs
plot(state, type.prob = "filtered") # filtered state probs
plot(state, type.prob = "viterbi") # fitted states
par(op)
```

The full output can be found in Appendix A.4.

4.2.4 Volatility Forecasts

It is possible to make conditional volatility forecasts with the generic function `predict()` by passing the `fit` object as can be seen below.

```
# one-step ahead prediction
predict(fit, nahead=1)$vol

## 2000-10-21
## 0.9340096
```

Note, we have yet to implement multi-step predictions for TV MSGARCH.

```
# multi-step ahead prediction
predict(fit, nahead=10)$vol
```

4.2.5 Predictive Densities & Risk Measures

We are able to make distributional forecasts with `PredPdf`, by creating a support mesh `x` which will be evaluated by the one-step-ahead predictive density. The one-step-ahead predictive density, as can be seen from the general result in [Equation 6](#), is

$$p(y_{T+1}|\Theta, \mathcal{F}_T) = \sum_{i=1}^K \sum_{j=1}^K P_{ij,t} \Pr(S_T = j|\Theta, \mathcal{F}_T) \mathcal{D}(y_{T+1}|S_{T+1} = i, \Theta, \mathcal{F}_T),$$

which can be evaluated by means of Hamilton's Filter ([J. D. Hamilton, 1989](#)).

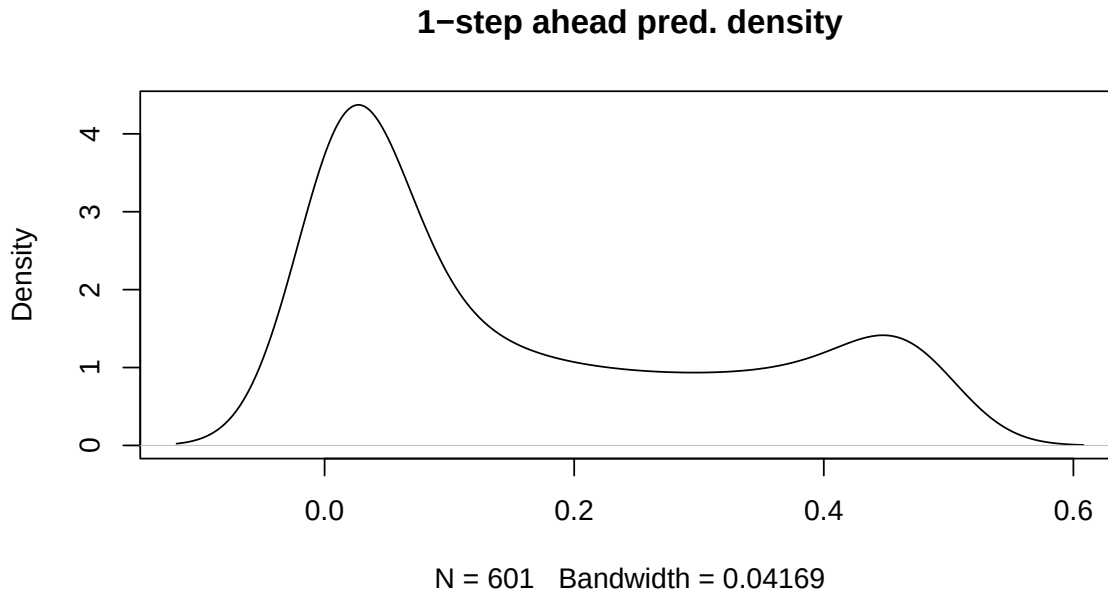
```
# create a mesh
x = seq(-3, 3, 0.01)

# one-step ahead predicitive density
pred.pdf = PredPdf(fit, do.its = FALSE, x = x) |> as.vector()

# create a density object
pred.density = density(pred.pdf)
```

Note, that we have yet to implement multi-step pdf predictions for TV MSGARCH.

```
# plot the density
plot(pred.density, main="1-step ahead pred. density")
```



By being able to conduct distributional forecasts, we can also estimate one-step ahead risk measures like VaR (Value-at-Risk) and ES (Expected Shortfall). This is done through numerically approximating the α quantile of one-step-ahead predictive distribution such that

$$VaR_{T+1}^\alpha = \inf\{y_{T+1} \in \mathbb{R} | F(y_{T+1} | \mathcal{F}_T) = \alpha\} \quad \text{and}$$

$$ES_{T+1}^\alpha = \frac{1}{\alpha} \int_{-\infty}^{VaR_{T+1}^\alpha} x f(x | \mathcal{F}_T) dx.$$

For the risk measure estimation one needs to supply the `fit` object to the `Risk()` functions.

```
Risk(fit, alpha=c(0.01, 0.05))

## $VaR
##           0.01      0.05
## 2000-10-21 -2.447991 -1.47649
##
## $ES
##           0.01      0.05
## 2000-10-21 -3.104105 -2.035367
```

5 Case Study on High Frequency Returns

Finally, the TV-MSGARCH model was applied to model the volatility of high-frequency equity returns to answer the research question: *"Do high-frequency (HF) returns have an effect on transition probabilities of volatility regimes and, if so, how do effects differ across the state space?"*. The relevance of such findings, on one hand, lies in finding out more about the statistical properties of intraday equity returns themselves. On the other hand, this modelling approach through TV-MSGARCH could provide a better performance for volatility or risk predictions in an intraday setting, which in turn could be employed by financial market participants, such as market makers or high-frequency traders, to obtain better methods of intraday risk assessment.

As indicated in the example in [subsubsection 4.2.2](#), by modelling transition probabilities in the TV-MSGARCH model through past returns, asymmetric effects on the probability of switching between states could be observed. The expected effect on transition probabilities could be observed through the estimated coefficients of past returns. The extended MSGARCH package and the time-varying switching capabilities rendered such estimations possible.

To answer how lagged returns can influence regime transition probabilities of intraday returns, we conduct the following experiment:

1. We collect high frequency price data for the *SP500 ETF* (SPY) for the entire month of March 2019.
2. We aggregate the data for different frequencies (1sec, 5sec, 10sec, 15sec, 30sec, 1min, 5min, 10min, 15min, 30min).
3. We relate transition probabilities to past returns i.e.

$$P_{t,lk} = \frac{\exp(\gamma_{0,lk} + \gamma_{1,lk}y_{t-1})}{1 + \exp(\gamma_{0,lk} + \gamma_{1,lk}y_{t-1})}.$$

4. We analyse the estimated coefficients in a 2-regime TV-MSGARCH model on a rolling basis (non-overlapping intervals).
5. We visualise the time series of coefficients.

5.1 Data

The high-frequency data is obtained from the *Lobster database* (<https://lobsterdata.com/>). We choose the SP500 ETF contract as a proxy for the US equity market. This does not mean that any results that are observed for the SP500 ETF can directly be applied to any individual stock or, in fact, to the equity market in general. However, given the time- and resource

constraints of this thesis, focusing on one particular time series is the most feasible solution that we can provide to answer the research question. Furthermore, we are constrained in the amount of data we can process throughout the duration of this thesis project. Hence, we reduce the amount of high-frequency data to one month of tick-data. We choose the year 2019 to exclude any kind of Covid-19 related effects. We choose the month of March due to it being the month with the highest trading volume in 2019. This is a desirable property as there is more intraday information and it decreases the risk of observing a day with periods in which no trading happens. This directly affects the aggregation of closing prices and returns as periods for which no trading information are available need to be imputed in order to construct an equidistant time series of prices/returns.

5.2 Preprocessing & Feature Engineering

In the following section we outline the data preprocessing step and how we create important features that are subsequently used in the TV-MSGARCH model to estimate transition probabilities.

In the preprocessing step the tick-data is transformed such that an equidistant time series of relevant order book features can be constructed. Tick-data includes all recorded messages and actions that occurred throughout the trading day on the exchange. After-hour trading activity is excluded as part of the preprocessing step. The tick-data contains information about the amount of limit orders, i.e. orders that are to be exercised at a particular price, for BID (buy) and ASK (sell) prices. For each side, BID and ASK, 10 levels of order book depth are retrieved, i.e. the best 10 BID and the best 10 ASK prices with the corresponding number of contracts (size) posted at the respective prices. Through order book information it was possible to create features that helped to estimate transition probabilities in the TV-MSGARCH model.

At first, given the tick-data, we have to choose a measure for the **price** of the time series. Choosing a measure for the underlying price is not straightforward. As a price proxy for order book data the volume-weighted *mid price*, as suggested by [Gatheral et al. \(2010\)](#), is computed over all 10 levels of order book depth. The volume-weighted mid price is computed as follows:

$$VweightedMidPrice = \frac{\sum_{i=1}^{10} BID_{i,Size} \times BID_{i,Price} + ASK_{i,Size} \times ASK_{i,Price}}{\sum_{i=1}^{10} ASK_{i,Size} + BID_{i,Size}} \quad (11)$$

The level 1 mid price, i.e. average between the best BID and the best ASK

$$Level1MidPrice = \frac{BID_{1,Price} + ASK_{1,Price}}{2},$$

which could be thought as a potential candidate for a price measure, poses problems when estimating the TV-MSGARCH with log-returns. For the level 1 mid price approach, the prices are much less volatile than for the volume-weighted mid price in [Equation 11](#). As the prices for the level 1 mid price move much more rarely than for the volume-weighted mid price, the log-returns are frequently equal to 0. That poses a problem for the EM and Viterbi algorithm in the ML estimation of the TV-MSGARCH model, since all zero-return observations are assigned to one regime – this regime has, accordingly, zero variance, which causes the algorithm to break down. Hence, the volume-weighted mid price is chosen as there is slightly more movement in the price process such that the log-returns would infrequently be exactly equal to zero. A snippet of the time series of log-returns aggregated for 1 second closing-prices is visualised in the two figures below.

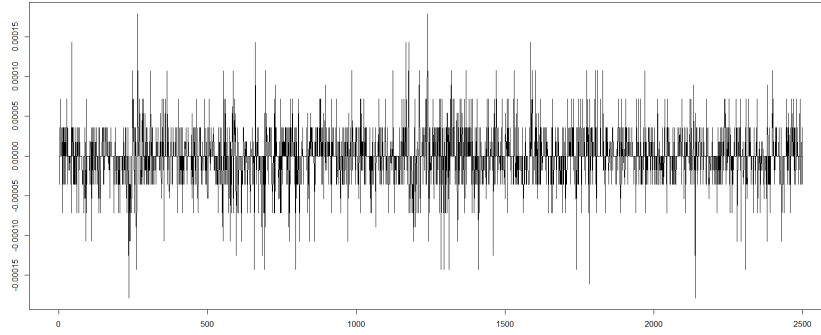


Figure 2: Illustration of midprice-based log-returns

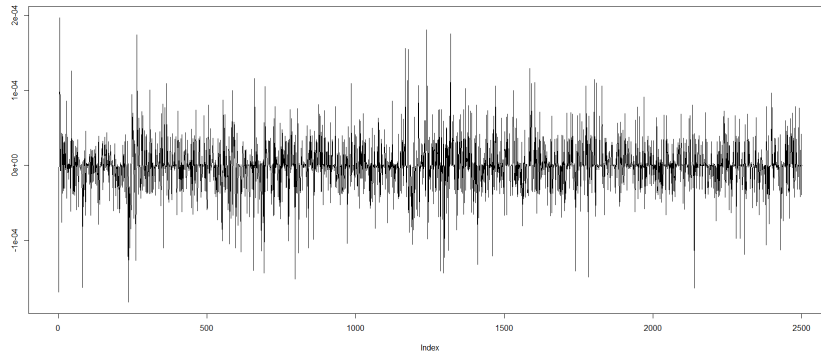


Figure 3: Illustration of volume-weighted-price-based log-returns

Part of the preprocessing is the aggregation of tick-data into the relevant frequencies (1sec, 5sec, 10sec, 15sec, 30sec, 1min, 5min, 10min, 15min, 30min). For the aggregation of prices, the closing prices for a given period are taken. For the aggregation of trading volumes at each BID/ASK level, the average sizes across the period are taken. In addition to the price estimate, the **spread**

$$Spread = ASK_{1,Price} - BID_{1,Price}$$

is computed and aggregated, i.e. averaged over the relevant period. To capture the variance of prices within a period corresponding to a particular frequency, the **quadratic variation**

$$[X]_t = \sum_{0 < s \leq t} (X_t - X_{t-})$$

is computed. The term X_t denotes the proxy for the price for tick-data at time t and X_{t-} denotes the price proxy at the last tick (recording of the order book). The variable is aggregated for all mentioned frequencies by summation, i.e. $[X]_t$ is summed up for the relevant period.

Ultimately, time series of **price**, **spread** and **quadratic variation** for aforementioned frequencies are constructed, based on the entire available data for March 2019. After aggregation the following additional features (based on the aggregated data) are computed:

Log-returns are computed based on the aggregated price data for each frequency. Given the aggregated price $\hat{X}_t$, log-returns are computed as $r_t = \log(\hat{X}_t) - \log(\hat{X}_{t-1})$.

The **density values** of log-returns are computed based on the aggregated price/return data. Analogous to [Haas et al. \(2013\)](#) who uses the likelihood value of returns to parameterise the time-varying mixing weights for Gaussian mixture models, we use the feature in our feature space to later on model transition probabilities for TV-MSGARCH models. This feature should capture the "likelihood", given the assumption of normal-distributed log-returns, of observing a particular log-return r_t . The mean of the distribution is assumed to be zero, while the variance is estimated as the sample variance of log-returns. Through incorporating it into the estimation of the transition probability, one can observe the potential effect that the likelihood of a particular observation might have on the transition probability, which was the main idea behind the inclusion of this feature.

Before we incorporate more features like **quadratic variation**, **spread** or **density values** in to the transition probability estimation of TV-MSGARCH, we model the transition probabilities, as previously mentioned, solely through lagged log-returns to answer the research question. Further experiments with a more complex feature space are discussed in [subsection 5.4](#).

5.3 Experiment to Answer the Research Question

After preprocessing the data and constructing the features to model transition probabilities, we test a set of models to answer the research question. For this purpose we model the transition probabilities through past log-returns:

$$P_{t,lk} = \frac{\exp(\gamma_{0,lk} + \gamma_{1,lk}y_{t-1})}{1 + \exp(\gamma_{0,lk} + \gamma_{1,lk}y_{t-1})}.$$

We fit TV-MSGARCH models by means of the extended MSGARCH package on a rolling basis for the entire data available. We try to achieve an approximate sample size of 2500 data points per estimation. Due to the aggregation for different frequencies, the number of estimations decreases with lower frequencies. We record and visualise the estimated coefficients $\gamma_{i,lk}$ for $i = 0, 1$ and $l \in \mathcal{K} = \{1, 2\}, k \in \mathcal{K} \setminus \{k_0\} = \{1\}$. The time-series of estimated coefficients can be seen in the figure below, in which we indicated the coefficients that are insignificant (at the 5 % level) with the color black. Coefficients that were significant are marked as red or green, depending on whether they are negative or positive, respectively. The plots also include information on what percentage of coefficients are estimated to be positive & significant and negative & significant (on top of each sub-figure).

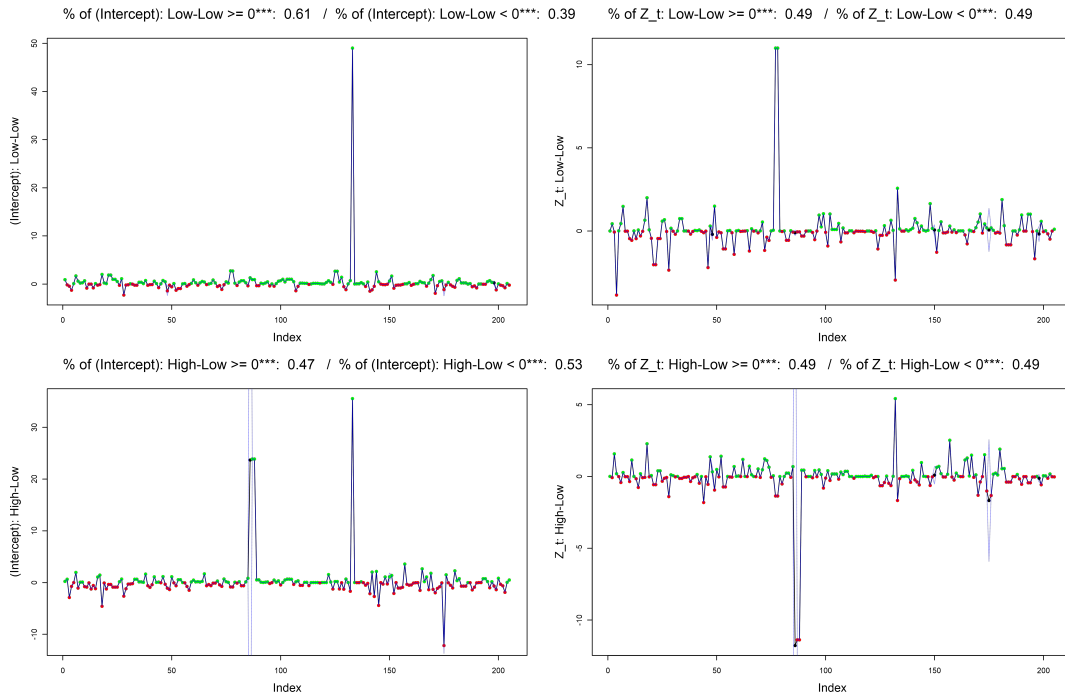


Figure 4: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 1 second frequency

The visualised time series of estimated coefficients for other frequencies are included in Ap-

pendix A.5. To obtain a clearer picture we average the coefficients of the rolling regressions for all frequencies:

Coefficient	1 Sec	5 Sec	10 Sec	15 Sec	30 Sec	1 Min
(Intercept) Low-Low	0.38	0.97	-1.62	1.58	2.34	3.57
(Z_t) Low-Low	0.06	0.12	2.68	0.14	0.48	0.46
(Intercept) High-Low	0.32	-0.18	0.83	2.43	15.71	14.46
(Z_t) High-Low	-0.12	0.52	2.43	2.09	12.90	24.09
# of estimations	205	41	20	13	6	3
	5 Min	10 Min	15 Min	30 Min		
(Intercept) Low-Low	3.13	2.66	4.73	4.3		
(Z_t) Low-Low	0.56	0.08	-0.72	-1.53		
(Intercept) High-Low	3.94	6.75	7.72	-223.		
(Z_t) High-Low	1.32	0.02	-0.99	-95.8		
# of estimations (observations)	1 (1713)	1 (857)	1 (571)	1 (286)		

Table 1: Average estimated coefficients from TV-MSGARCH for multiple frequencies

From the obtained time series of coefficients for $\gamma_{i,lk}$ we can observe that

- significance seems to decrease with lower frequencies and
- lagged returns seem to have mostly a positive effect on the transition probabilities $P_{Low,Low}$ and $P_{High,High}$ ¹¹.

Given a positive coefficient for lagged returns, the effect on probabilities can be summarised as follows:

	t+1 Low	t+1 High
t Low	+	-
t High	+	-

Table 2: Effect of increase of lagged return on probabilities

Essentially, on average the probability of transitioning into a high volatility state decreases with positive returns and increases with negative returns, regardless of whether the time series was in a low- or high volatility regime one period before the transition. This would provide evidence in support of the *leverage effect* (Black, 1976) as high-volatility regimes are associated with negative returns. However, the coefficients are not always positive and significant. The evidence for such an effect reflected in the transition probabilities of a TV-MSGARCH model is therefore limited. Part of the problem is the time- and data- constraints of this master thesis.

¹¹Whether a state in the estimation in MSGARCH is considered to be a low or high volatility state is determined by the conditional volatility of each regime. The output is always ordered according to the unconditional regime-specific volatility.

Perhaps with more data and more available processing time a more thorough examination of coefficients $\gamma_{i,lk}$ could have been conducted. The available tick-data (especially for lower frequencies) posed a problem as merely a single estimation could sometimes be done as seen from [Figure 5.3](#).

5.4 Out-of-Sample Performance

As we see in [subsubsection 4.2.2](#), the in-sample fit for TV-MSGARCH models can be on par with the in-sample fit of MSGARCH models (if not even better). However, the actual applicability or usefulness of the TV-MSGARCH model should not only be assessed by its in-sample fit, but more so by its forecasting performance as this is the main interest for industry practitioners who use GARCH-type models on a daily basis for risk and volatility forecasting purposes. Hence, we carry out the following experiment:

- We model the transition probabilities using the following features:
 - lag-1 log-returns,
 - lag-1 quadratic variation,
 - lag-1 spread,
 - lag-1 density of log-returns.
- We conduct a rolling VaR (Value-at-Risk) prediction for all frequencies at the 1%- and 5% level.
- We compare 3 models: TV MSGARCH, MSGARCH and single-regime GARCH.
- We compare the models above across skewed and symmetric Student-t distributions.
- We report the *average quantile loss* for the VaR Risk predictions.
- We use the same data as in the previous section.

To assess the performance of a particular GARCH model, we use the quantile loss method by [González-Rivera et al. \(2004\)](#), which is implemented in the GAS packages by [Ardia et al. \(2016\)](#). The quantile loss is given by

$$QL_t(\alpha) = (\alpha - \mathbf{1}_{\{y_t < VaR_t(\alpha)\}})(y_t - VaR_t(\alpha)).$$

It is an asymmetric loss function that punishes VaR-exceedences (i.e. one-step-ahead predictions VaR_t , which satisfy $y_t < VaR_t$) heavily by a weight of $1 - \alpha$. Quantile loss values are averaged over the backtest period to give an average $QL(\alpha)$ value. The measure is minimal

(equal to 0) when the α quantiles are perfectly predicted over the entire backtest period. Finally, the model with the smallest $QL(\alpha)$ value is deemed to be better.

In the table below we provide the average quantile loss for various GARCH model specifications. The log-returns in the estimation are scaled by a factor of 100 for better readability. Note, `in_sample` denotes the in-sample size on which the model is fitted, `out_sample` denotes the number of out-of-sample observations and `k_skip` denotes the number of observations that are skipped along the rolling estimation (this has to be done to decrease the number of estimations and to cover the entire available data at the same time).

Freqs.	STD			SSTD			in_sample	out_sample	k_skip
	TV MSGARCH	MSGARCH	GARCH	TV MSGARCH	MSGARCH	GARCH			
VaR 5%									
1sec	11.68	11.58	12.69	11.87	11.47	12.72	1500	1629	300
5sec	10.56	9.75	9.8	10.37	9.78	9.81	1500	1379	70
10sec	12.34	10.29	10.24	11.19	10.4	10.23	1500	1357	35
15sec	11.37	9.66	9.76	10.47	9.67	9.76	1500	1246	25
30sec	10.92	9.88	9.82	10.58	9.87	9.84	1500	1233	12
1min	11.11	10.4	10.48	11.23	10.52	10.5	1500	1326	5
5min	13.11	13.24	13.19	13.32	13.29	13.13	1000	593	1
10min	11.4	11.13	10.97	10.98	11.08	10.87	600	176	1
15min	13.17	12.27	12.38	13.08	12.87	12.4	400	143	1
30min	15.68	12.93	12.87	14.13	12.18	12.78	200	71	1
VaR 1%									
1sec	3.94	3.77	4.56	4.11	3.69	4.65	1500	1629	300
5sec	3.3	2.72	2.75	2.97	2.75	2.75	1500	1379	70
10sec	4.25	3.12	3.34	3.67	3.39	3.36	1500	1357	35
15sec	3.59	2.76	2.74	3.3	2.75	2.75	1500	1246	25
30sec	4.11	3.1	3.06	3.78	3.05	3.05	1500	1233	12
1min	3.52	3.06	3.11	4.12	3.16	3.11	1500	1326	5
5min	3.96	3.99	3.93	4.14	4.07	3.97	1000	593	1
10min	3.09	2.76	2.64	2.99	2.72	2.74	600	176	1
15min	3.46	3.46	3.23	3.87	3.55	3.23	400	143	1
30min	3.86	2.8	2.86	3.91	2.91	2.92	200	71	1

Table 3: Average Quantile Loss for log-returns $\times 100$ for multiple model specifications

First, we compare the results in Table 3 by different VaR quantiles. Note that it is logical that the 1% VaR predictions exhibit lower average quantile loss values than the 5% predictions since there are naturally less observations that fall below these quantiles and thus the loss function does not get to punish as many VaR-exceedences as in the 5% quantile case.

Second, we can observe no noticeable difference in the choice of the distribution. Skewed or non-skewed versions of the Student-t distribution seem to perform equally well as the out-performance values within the model specifications TV-MSGARCH, MSGARCH and GARCH are rather balanced across different frequencies.

Third, we compare the model performance within a particular distribution. It can be seen that

TV-MSGARCH only outperformed the other models for the 5 min frequency within the symmetric distribution category. Within the skewed distribution category the TV-MSGARCH model was strictly outperformed by MSGARCH and single-regime models. This underwhelming result for the TV-MSGARCH model speaks for the fact that this model is most probably not the best choice when it comes to modelling intraday volatility or risk. Perhaps high-frequency returns are too noisy to incorporate external regime-switching probabilities as different regimes are indistinguishable from each other. The over-fitting factor that is reflected in the high amount of parameters to estimate in the TV-MSGARCH model, compared to the competitor models, could serve as another explanation.

In the figure below one can see a visual representation of the VaR predictions. For illustrative purposes, we plot the predictions and the respective observations in the 5 min frequency (as this constitutes the best relative performance of the TV-MSGARCH model).

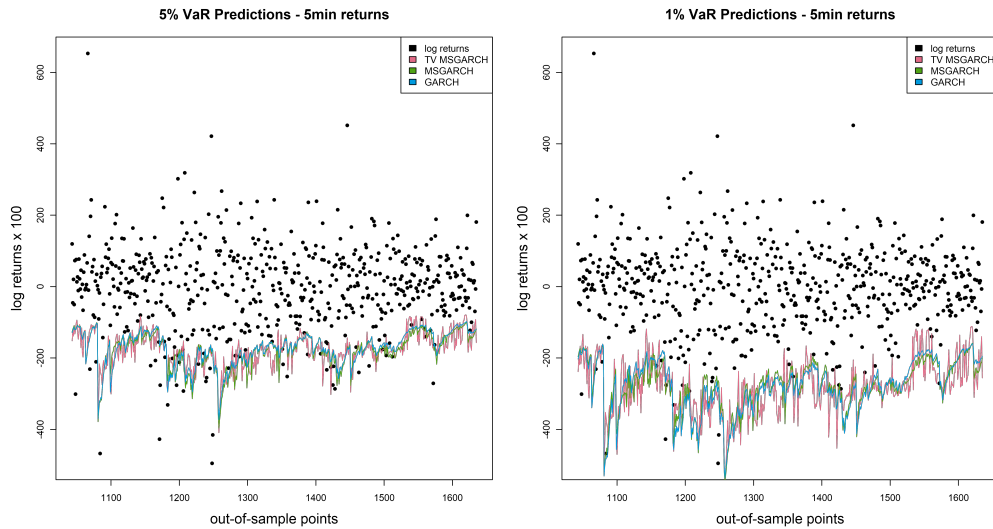


Figure 5: VaR prediction performance comparison for symmetric Student-t distributed innovations - 5 minute frequency

By visual inspection one can see that VaR predictions for all three models (TV-MSGARCH, MSGARCH and single-regime GARCH) are rather similar. The VaR predictions for the TV-MSGARCH model itself seem to be more volatile. This could be due to the fact that the returns, which feed into the transition probabilities, are volatile themselves, which in turn causes the predictions to be more erratic. However, this phenomenon was not investigated further.

5.5 Applying TV-MSGARCH to Daily Returns

Since the intraday results for the more complex¹² TV-MSGARCH model are rather underwhelming, we apply the TV-MSGARCH model to daily returns. Hence, we retrieve daily price data for the SP500 ETF (SPY) for the last 12 years (2010 to 2022). For the retrieved daily price data there is no access to intraday order book information. Thus, we employed the previous approach, as in [subsection 5.3](#), and related switching probabilities solely to lagged returns.

In the experiment, a comparison across three dimensions is conducted:

- We compare the performance between TV-MSGARCH, MSGARCH and single-regime model specifications.
- We compare the performance between multiple distributions (skewed and symmetric).
- We compare the performance between symmetric and asymmetric¹³ GARCH specifications (GARCH and GJR GARCH).

This experiment constitutes an analog to [Ardia \(2017\)](#), who compared MSGARCH and single-regime GARCH models across the same dimensions. The difference to our approach is that we focus only on one particular time series (SP500 ETF - SPY) rather than multiple stocks, while also including the TV-MSGARCH model.

The out-of-sample performance, as in the experiment in the [subsection 5.4](#), is summarised in [Table 4](#) through the average quantile loss. Note that in this experiment the log-returns are not scaled by a factor of 100, which explains why the average quantile loss values are much smaller than in the previous experiment.

Distr.	GARCH			GJR			In_sample	Out_sample	k_skip
	TV-MSGARCH	MSGARCH	GARCH	TV-MSGJR	MSGJR	GJR			
VaR 5%									
norm	0.1306	0.1312	0.1295	0.1305	0.1311	0.1305	2000	1020	1
snorm	0.1298	0.1297	0.1292	0.1260	0.1268	0.1266	2000	1020	1
std	0.1312	0.1315	0.1311	0.1275	0.1274	0.1276	2000	1020	1
sstd	0.1305	0.1311	0.1305	0.1279	0.1264	0.1270	2000	1020	1
VaR 1%									
norm	0.0452	0.0463	0.0478	0.0432	0.0449	0.0447	2000	1020	1
snorm	0.0436	0.0446	0.0459	0.0427	0.0431	0.0448	2000	1020	1
std	0.0461	0.0457	0.0457	0.0456	0.0438	0.0448	2000	1020	1
sstd	0.0432	0.0449	0.0447	0.0447	0.0415	0.0441	2000	1020	1

Table 4: Average quantile loss - a comparison of GARCH model specifications for daily returns (2010-2022)

¹²Complex here refers to the number of features that were utilised to model transition probabilities.

¹³Symmetric and asymmetric refer to the effect of returns on to the variance process.

First, among the category of normal-distributed innovations, we observe that skewed distributions seem to outperform symmetric distributions across all models. Among Student-t-distributed innovations, skewed distributions seem to outperform non-skewed distributions in most of the models. Thus, it can be concluded that skewed distributions are preferred when modelling volatility of daily log-returns for the SP500 through a GARCH or GJR GARCH model.

Second, when we compare the group of normal distributions with the group of Student-t distributions, then we see that normal-distributed innovations are preferred at the 5% VaR level, whereas Student-t distributed innovations are slightly preferred at the 1% level.

Third, when comparing the performance among symmetric and asymmetric GARCH specifications, i.e. GARCH versus GJR in [Table 4](#), we observe that the asymmetric specification outperforms the symmetric version almost in every case across all three GARCH models.

Fourth, when comparing TV-MSGARCH, MSGARCH and single-regime GARCH models, we observe that the TV-MSGARCH model within the group of 'GARCH' models (i.e. the first 3 columns in [Table 4](#)) outperforms the other two models in 5 out of 8 cases. Within the group of 'GJR' models (i.e. columns 4-6 in [Table 4](#)) TV-MSGARCH outperforms the other two models in 4 out of 8 times. This constitutes a much better performance for the TV-MSGARCH model as for high-frequency returns. Ultimately, the best model for the given data was the Markov switching GJR GARCH model. However, the aforementioned results present the TV-MSGARCH as a much more promising model class as the experiments in [subsection 5.4](#) suggest. This gives reason to believe that time-varying regime switching is theoretically able to provide better risk forecasts in certain scenarios.

In this experiment we related regime switching probabilities to past returns only. Given that regimes in financial markets are often driven by economic indicators like inflation, interest-rates or geo-political events, one might be able to construct even better models by incorporating these economic indicators into the TV-MSGARCH model.

6 Conclusion

We found evidence of asymmetric effects on transition probabilities of volatility regimes within intraday returns. By relating transition probabilities to past return, the leverage effect ([Black, 1976](#)) could be observed in the estimated coefficients of returns when volatility was modelled through a time-varying Markov switching GARCH model. According to our findings, negative high-frequency returns seem to be associated with high volatility regimes.

The R package MSGARCH by [Ardia et al. \(2019\)](#) was extended to incorporate time-varying transition probabilities. The package extension is freely available online for anyone to use. A walk-through of capabilities that are supported for time-varying switching was provided as part of this thesis.

Subsequently, the model implementation was applied to high-frequency returns and daily returns. The out-of-sample performance of various models was compared by means of Value-at-Risk predictions. There was no evidence that TV-MSGARCH models perform better than MSGARCH or single-regime models in a setting where high-frequency returns are considered for the SP500 ETF (SPY). However, there was some evidence that TV-MSGARCH can, in fact, outperform other model specifications for the SP500 ETF (SPY) when daily return data is considered. Nevertheless, if the model performance was to also consider the time it takes to estimate such models, then the difference between previously mention model specifications might be negligible as it was observed (even though not explicitly recorded) that TV-MSGARCH models take much longer to fit as other competitor models.

Ultimately, the TV-MSGARCH model (in theory) represents a flexible model class that allows the incorporation of exogenous variates to gain more information of regime transitions. These models can possibly be improved in the future by including covariates which contain more explanatory power with respect to the transition between volatility regimes.

References

- Ardia, David (Oct. 2017). R Finance 2017. URL: <https://youtu.be/yRo2fnDS2J4>.
- Ardia, David, Keven Bluteau, Kris Boudt, Leopoldo Catania, and Denis-Alexandre Trottier (2019). “Markov-switching GARCH models in R: The MSGARCH package”. *Journal of Statistical Software* 91.4.
- Ardia, David, Kris Boudt, and Leopoldo Catania (2016). “Generalized autoregressive score models in R: The GAS package”. *arXiv preprint arXiv:1609.02354*.
- Baum, Leonard E, Ted Petrie, George Soules, and Norman Weiss (1970). “A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains”. *The annals of mathematical statistics* 41.1, pp. 164–171.
- Black, Fischer (1976). “Studies of stock market volatility changes”. *1976 Proceedings of the American statistical association business and economic statistics section*.
- Bollerslev, Tim (1986). “Generalized autoregressive conditional heteroskedasticity”. *Journal of econometrics* 31.3, pp. 307–327.
- Broyden, Charles George (1970). “The convergence of a class of double-rank minimization algorithms 1. general considerations”. *IMA Journal of Applied Mathematics* 6.1, pp. 76–90.
- Brunetti, Celso, Chiara Scotti, Roberto S Mariano, and Augustine HH Tan (2008). “Markov switching GARCH models of currency turmoil in Southeast Asia”. *Emerging Markets Review* 9.2, pp. 104–128.
- Cai, Jun (1994). “A Markov model of switching-regime ARCH”. *Journal of Business & Economic Statistics* 12.3, pp. 309–316.
- Dempster, Arthur P, Nan M Laird, and Donald B Rubin (1977). “Maximum likelihood from incomplete data via the EM algorithm”. *Journal of the Royal Statistical Society: Series B (Methodological)* 39.1, pp. 1–22.
- Dewachter, Hans (2001). “Can Markov switching models replicate chartist profits in the foreign exchange market?” *Journal of International Money and finance* 20.1, pp. 25–41.
- Diebold, Francis X, Joon-Haeng Lee, and Gretchen C Weinbach (1994). “Regime switching with time-varying transition probabilities”. *Business Cycles: Durations, Dynamics, and Forecasting* 1, pp. 144–165.
- Duprey, Thibaut and Benjamin Klaus (2017). “How to predict financial stress? An assessment of Markov switching models”.
- Eddelbuettel, Dirk and Romain François (2011). “Rcpp: Seamless R and C++ Integration”. *Journal of Statistical Software* 40.8, pp. 1–18. DOI: [10.18637/jss.v040.i08](https://doi.org/10.18637/jss.v040.i08).
- (2022). “Exposing C++ functions and classes with Rcpp modules”. *Vignette included in R package Rcpp*, URL <https://CRAN.R-Project.org/package=Rcpp>.

- Eddelbuettel, Dirk and Conrad Sanderson (Mar. 2014). "RcppArmadillo: Accelerating R with high-performance C++ linear algebra". *Computational Statistics and Data Analysis* 71, pp. 1054–1063. URL: <http://dx.doi.org/10.1016/j.csda.2013.02.005>.
- Engel, Charles (1994). "Can the Markov switching model forecast exchange rates?" *Journal of international economics* 36.1-2, pp. 151–165.
- Engle, Charles and James D Hamilton (1990). "Long swings in the dollar: are they in the data and do markets know it". *American Economic Review* 80.4, pp. 689–713.
- Engle, Robert F (1982). "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation". *Econometrica: Journal of the econometric society*, pp. 987–1007.
- Engle, Robert F and Andrew J Patton (2007). "What good is a volatility model?" *Forecasting volatility in the financial markets*. Elsevier, pp. 47–63.
- Fernández, Carmen and Mark FJ Steel (1998). "On Bayesian modeling of fat tails and skewness". *Journal of the american statistical association* 93.441, pp. 359–371.
- Filardo, Andrew J (1994). "Business-cycle phases and their transitional dynamics". *Journal of Business & Economic Statistics* 12.3, pp. 299–308.
- Fornari, Fabio and Antonio Mele (1997). "Sign-and volatility-switching ARCH models: theory and applications to international stock markets". *Journal of Applied Econometrics* 12.1, pp. 49–65.
- Frühwirth-Schnatter, Sylvia (2006). *Finite mixture and Markov switching models*. Vol. 796. New York: Springer.
- Frühwirth-Schnatter, Sylvia, Gilles Celeux, and Christian P Robert (2019). *Handbook of mixture analysis*. CRC press.
- Gatheral, Jim and Roel CA Oomen (2010). "Zero-intelligence realized variance estimation". *Finance and Stochastics* 14.2, pp. 249–283.
- Glosten, Lawrence R, Ravi Jagannathan, and David E Runkle (1993). "On the relation between the expected value and the volatility of the nominal excess return on stocks". *The journal of finance* 48.5, pp. 1779–1801.
- González-Rivera, Gloria, Tae-Hwy Lee, and Santosh Mishra (2004). "Forecasting volatility: A reality check based on option pricing, utility function, value-at-risk, and predictive likelihood". *International Journal of forecasting* 20.4, pp. 629–645.
- Gray, Stephen F (1996). "Modeling the conditional distribution of interest rates as a regime-switching process". *Journal of Financial Economics* 42.1, pp. 27–62.
- Guidolin, Massimo (2011). "Markov switching models in empirical finance". *Missing data methods: Time-series methods and applications*. Emerald Group Publishing Limited.
- Haas, Markus, Jochen Krause, Marc S Paoletta, and Sven C Steude (2013). "Time-varying mixture GARCH models and asymmetric volatility". *The North American Journal of Economics and Finance* 26, pp. 602–623.

- Haas, Markus, Stefan Mittnik, and Marc S Paoletta (2004). "A new approach to Markov-switching GARCH models". *Journal of financial Econometrics* 2.4, pp. 493–530.
- Hamilton, J. (2020). *Time series analysis*. Princeton university press.
- Hamilton, James D (1988). "Rational-expectations econometric analysis of changes in regime: An investigation of the term structure of interest rates". *Journal of Economic Dynamics and Control* 12.2-3, pp. 385–423.
- (1989). "A new approach to the economic analysis of nonstationary time series and the business cycle". *Econometrica: Journal of the econometric society*, pp. 357–384.
- Hamilton, James D and Raul Susmel (1994). "Autoregressive conditional heteroskedasticity and changes in regime". *Journal of econometrics* 64.1-2, pp. 307–333.
- Klaassen, Franc (2002). "Improving GARCH volatility forecasts with regime-switching GARCH". *Advances in Markov-switching models*. Springer, pp. 223–254.
- Lamoureux, Christopher G and William D Lastrapes (1990). "Persistence in variance, structural change, and the GARCH model". *Journal of Business & Economic Statistics* 8.2, pp. 225–234.
- Maheu, John M and Thomas H McCurdy (2000). "Identifying bull and bear markets in stock returns". *Journal of Business & Economic Statistics* 18.1, pp. 100–112.
- McNeil, Alexander J, Rüdiger Frey, and Paul Embrechts (2015). *Quantitative risk management: concepts, techniques and tools-revised edition*. Princeton university press.
- Nelson, Daniel B (1991). "Conditional heteroskedasticity in asset returns: A new approach". *Econometrica: Journal of the econometric society*, pp. 347–370.
- Pagan, Adrian R and G William Schwert (1990). "Alternative models for conditional stock volatility". *Journal of econometrics* 45.1-2, pp. 267–290.
- Rabiner, Lawrence R (1989). "A tutorial on hidden Markov models and selected applications in speech recognition". *Proceedings of the IEEE* 77.2, pp. 257–286.
- Rydén, Tobias, Timo Teräsvirta, and Stefan Åsbrink (1998). "Stylized facts of daily return series and the hidden Markov model". *Journal of applied econometrics* 13.3, pp. 217–244.
- Schaller, Huntley and Simon Van Norden (1997). "Regime switching in stock market returns". *Applied Financial Economics* 7.2, pp. 177–191.
- Tan, Chia-Yen, You-Beng Koh, Kok-Haur Ng, and Kooi-Huat Ng (2021). "Dynamic volatility modelling of Bitcoin using time-varying transition probability Markov-switching GARCH model". *The North American Journal of Economics and Finance* 56, p. 101377.
- Trottier, Denis-Alexandre and David Ardia (2016). "Moments of standardized Fernandez–Steel skewed distributions: Applications to the estimation of GARCH-type models". *Finance Research Letters* 18, pp. 311–316.
- Viterbi, Andrew (1967). "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm". *IEEE transactions on Information Theory* 13.2, pp. 260–269.

Wang, Lu, Feng Ma, Jianyang Hao, and Xinxin Gao (2021). “Forecasting crude oil volatility with geopolitical risk: Do time-varying switching probabilities play a role?” *International Review of Financial Analysis* 76, p. 101756.

Zakoian, Jean-Michel (1994). “Threshold heteroskedastic models”. *Journal of Economic Dynamics and control* 18.5, pp. 931–955.

A Appendix

A.1

$$\Pr(S_t = k \mid \mathbf{P}) = \sum_{i=1}^K P_{ik} \Pr(S_{t-1} = i \mid \mathbf{P}) = \sum_{i=1}^K P_{ik} \pi_i = \pi_k \quad (12)$$

A.2

Assuming that all eigenvalues of $\text{diag}(\beta_1, \dots, \beta_K)$, i.e. $\beta_1, \dots, \beta_K$ are smaller than one in absolute value, the GARCH(1,1) process according to [Haas et al., 2004](#) can be expressed as an ARCH(∞) process:

$$\begin{aligned} h_t &= \alpha_0 + \alpha_1 y_{t-1}^2 + \text{diag}(\beta_1, \dots, \beta_K) h_{t-1} \\ &= \alpha_0 + \alpha_1 y_{t-1}^2 + \text{diag}(\beta_1, \dots, \beta_K) (\alpha_0 + \alpha_1 y_{t-2}^2 + \text{diag}(\beta_1, \dots, \beta_K) h_{t-2}) \\ &= \alpha_0 + \alpha_1 y_{t-1}^2 + \text{diag}(\beta_1, \dots, \beta_K) h_{t-1} \text{diag}(\beta_1, \dots, \beta_K) \alpha_0 + \text{diag}(\beta_1, \dots, \beta_K) \alpha_1 y_{t-2}^2 + \text{diag}(\beta_1, \dots, \beta_K)^2 h_{t-2} \\ &= \dots \\ &= [I + \text{diag}(\beta_1, \dots, \beta_K) + \text{diag}(\beta_1, \dots, \beta_K)^2 + \dots] \alpha_0 + \sum_{i=1}^{\infty} \text{diag}(\beta_1, \dots, \beta_K)^{i-1} \alpha_1 y_{t-i}^2 \\ &= [I - \text{diag}(\beta_1, \dots, \beta_K)]^{-1} \alpha_0 + \sum_{i=1}^{\infty} \text{diag}(\beta_1, \dots, \beta_K)^{i-1} \alpha_1 y_{t-i}^2 \end{aligned} \quad (13)$$

A.3

```
# TVMSGARCH for SMI data set
spec = CreateSpec(switch.spec = list(do.tvp=TRUE),
                  variance.spec = list(model = c("sGARCH", "sGARCH")),
                  distribution.spec = list(distribution = c("norm", "norm")))
# covariate matrix to model transition probabilities
Z = matrix(1, nrow = length(data)-1, ncol = 2)
Z[,2] = data[1:2499]
data = data[2:2500]
```

```
# fit the TVMSGARCH model via MLE
fit = FitML(spec = spec, data = data, Z = Z)

# MSGARCH for SMI data set
spec = CreateSpec(variance.spec = list(model = c("sGARCH", "sGARCH")),
                  distribution.spec = list(distribution = c("norm", "norm")))

fit.basic = FitML(spec, data)

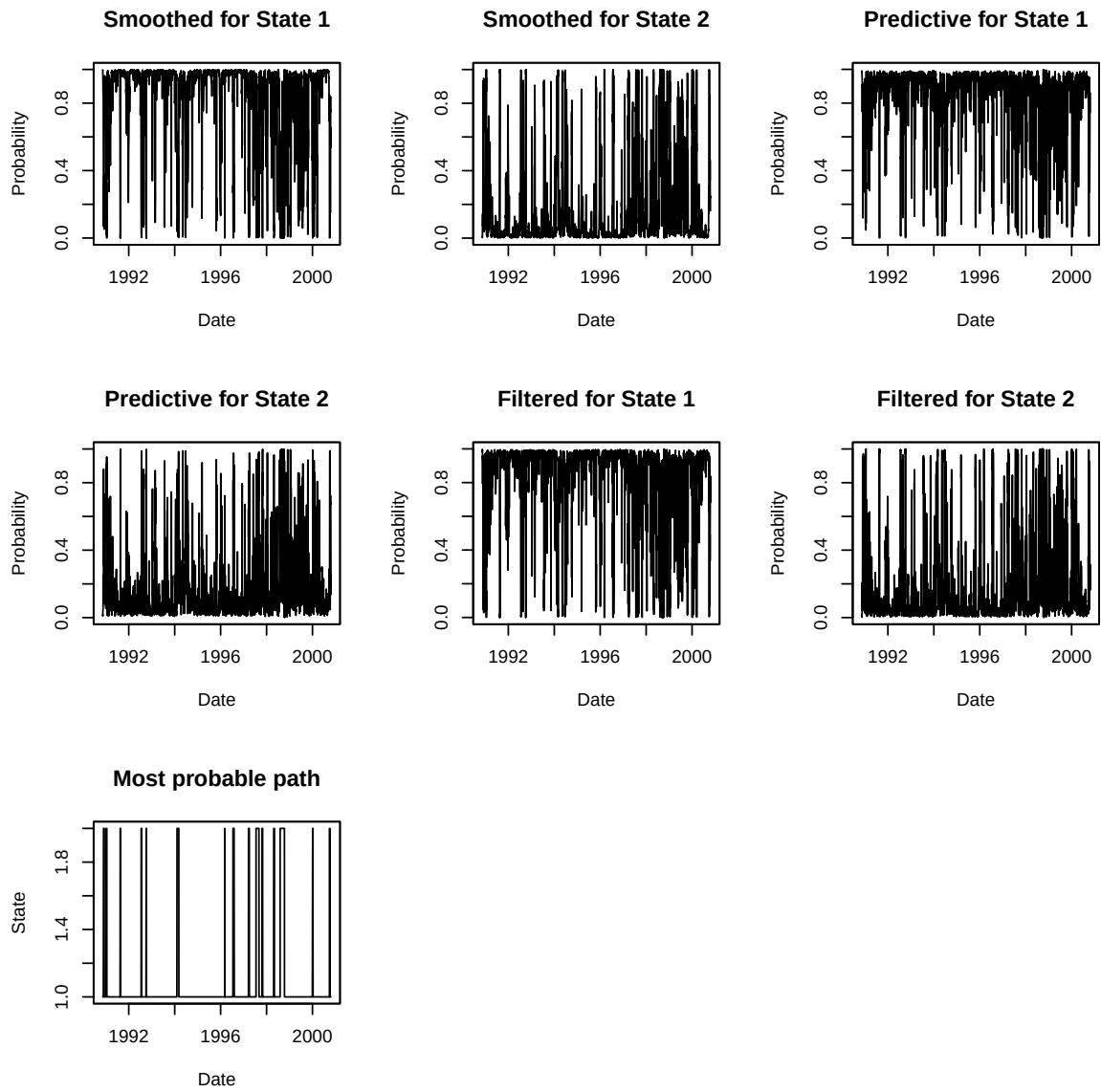
# summary
summary(fit)
summary(basic.fit)
```

```
print(fit)
#> Specification type: TV Markov-switching
#> Specification name: sGARCH_norm sGARCH_norm
#> Number of parameters in each variance model: 3 3
#> Number of parameters in each distribution: 0 0
#> -----
#> Fixed parameters:
#> None
#> -----
#> Across regime constrained parameters:
#> None
#> -----
#> Fitted parameters:
#>      Estimate Std. Error  t value Pr(>|t|)
#> alpha0_1      0.0060     0.0021   2.8446 2.223e-03
#> alpha1_1      0.0144     0.0074   1.9610 2.494e-02
#> beta_1        0.9691     0.0042 230.2402 <1e-16
#> alpha0_2      0.2023     0.1132   1.7871 3.696e-02
#> alpha1_2      0.0599     0.0657   0.9129 1.806e-01
#> beta_2        0.8997     0.0320 28.1478 <1e-16
#> factor0_11     4.0432     0.4458   9.0703 <1e-16
#> factor1_11     2.0588     0.3785   5.4400 2.664e-08
#> factor0_21    -1.7192     0.5571  -3.0862 1.014e-03
#> factor1_21     1.0324     0.4235   2.4381 7.382e-03
#> P_1_1          0.9965     3.3036   0.3017 3.815e-01
#> P_2_1          0.2202     0.0747   2.9474 1.602e-03
#> -----
#> Transition matrix:
#>      t=1/k=1 t=1/k=2
#> t=0/k=1  0.9965  0.0035
#> t=0/k=2  0.2202  0.7798
#> -----
#> LL: -3373.8684
#> AIC: 6771.7368
#> BIC: 6841.6205
#> -----
```

```
print(fit.basic)
#> Specification type: Markov-switching
#> Specification name: sGARCH_norm sGARCH_norm
#> Number of parameters in each variance model: 3 3
#> Number of parameters in each distribution: 0 0
#> -----
#> Fixed parameters:
#> None
#> -----
#> Across regime constrained parameters:
#> None
#> -----
#> Fitted parameters:
#>      Estimate Std. Error  t value Pr(>|t|)
#> alpha0_1      0.0221     0.0073   3.0305 1.221e-03
#> alpha1_1      0.0881     0.0343   2.5671 5.128e-03
#> beta_1        0.8802     0.0097  91.1006 <1e-16
#> alpha0_2      0.0207     0.0178   1.1636 1.223e-01
#> alpha1_2      0.0054     0.0061   0.8803 1.894e-01
#> beta_2        0.9940     0.0005 2020.8306 <1e-16
#> P_1_1          0.9785     0.0002 6047.5704 <1e-16
#> P_2_1          0.9997     0.9150   1.0925 1.373e-01
#> -----
#> Transition matrix:
#>      t+1/k=1 t+1/k=2
#> t/k=1  0.9785  0.0215
#> t/k=2  0.9997  0.0003
#> -----
#> Stable probabilities:
#> State 1 State 2
#> 0.979  0.021
#> -----
#> LL: -3388.461
#> AIC: 6792.922
#> BIC: 6839.5111
#> -----
```

Figure 6: A comparison of outputs - TVMSGARCH vs. MSGARCH - Data: SMI

A.4



A.5

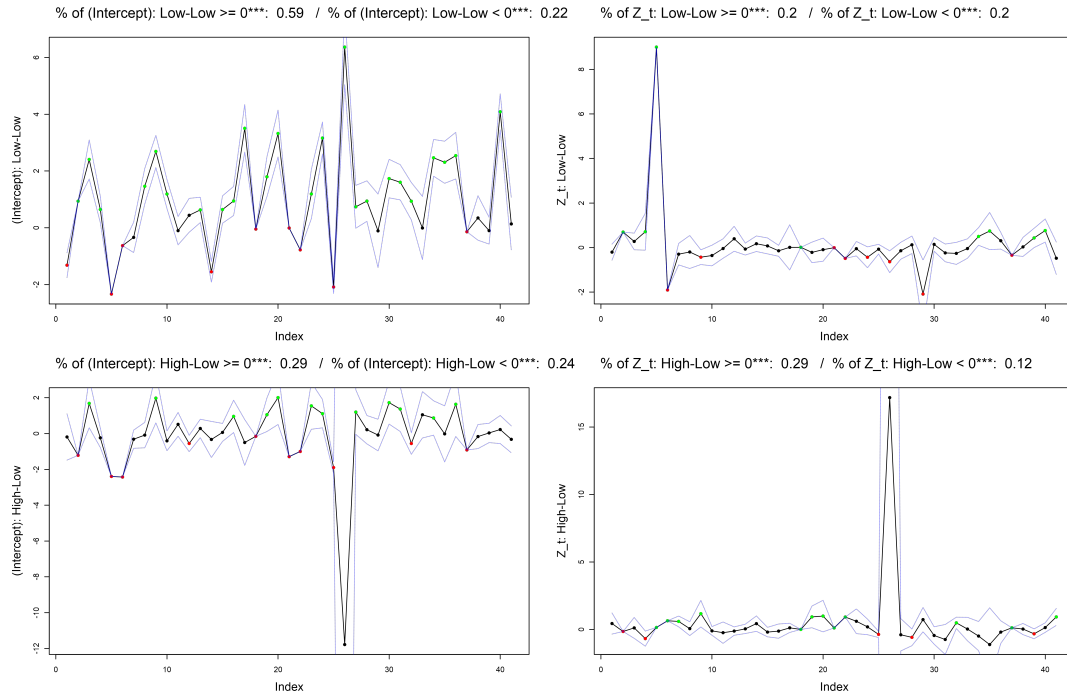


Figure 7: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 5 second frequency

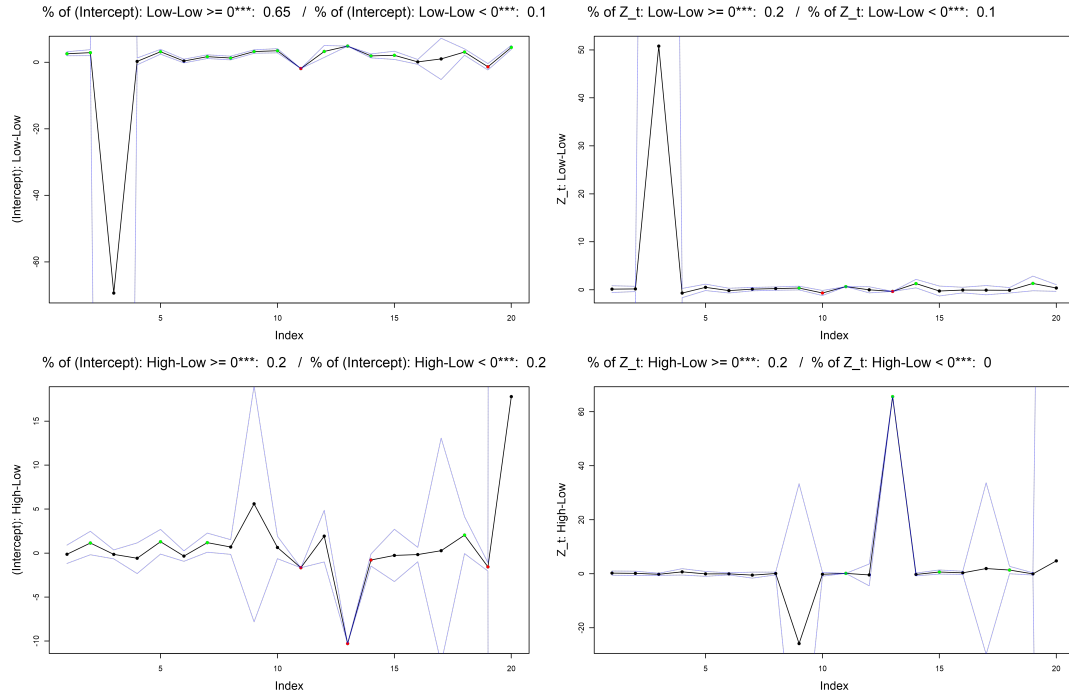


Figure 8: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 10 second frequency

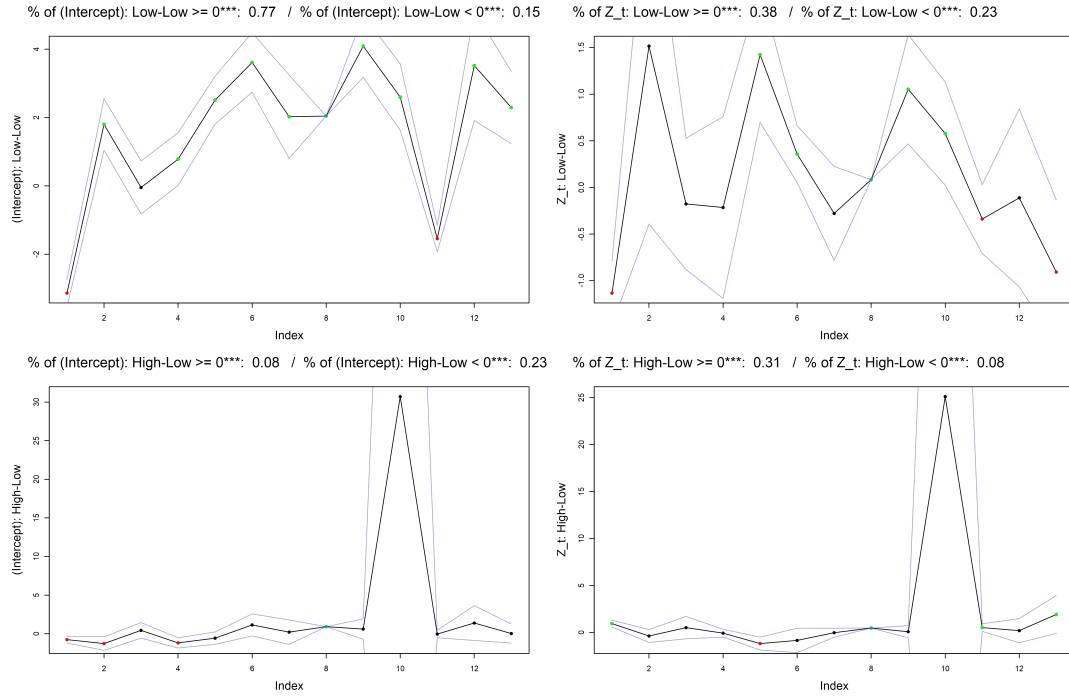


Figure 9: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 15 second frequency

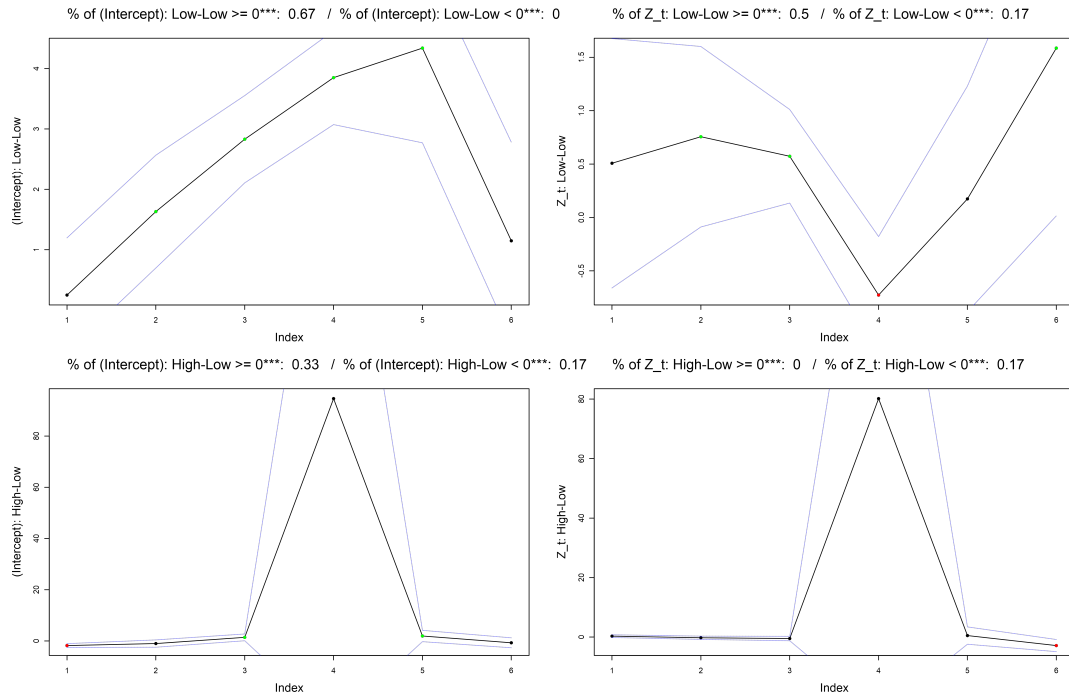


Figure 10: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 30 second frequency

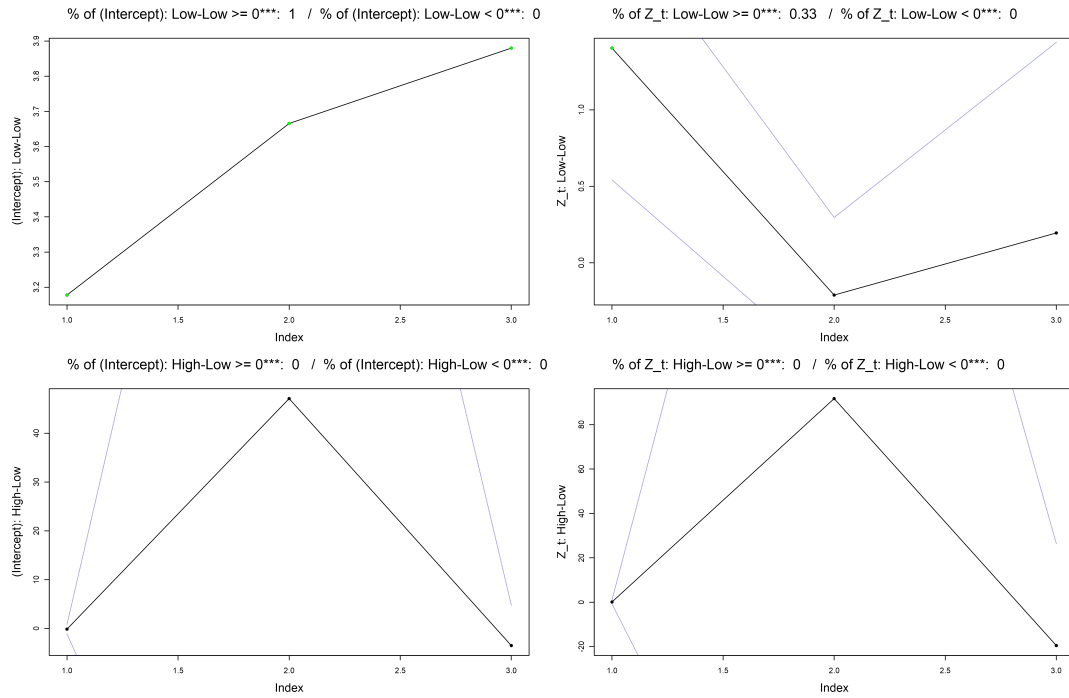


Figure 11: Time series of estimated coefficients in the TV-MSGARCH model - based on lagged log-returns - 1 minute frequency